

Lahore Wastewater and Drainage Management Section 6 – Employer Requirements (ER)
Project (LWDMP) – Sewerage System from Larechs
Colony to Gulshan-e-Ravi Lahore (Through
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	• Lahore Waste Management Company (LWMC) • Punjab Irrigation Department (PID) Facilitation for Site Inspection The Contractor shall allow these entities to enter and inspect the site as required to ensure smooth coordination and compliance with project objectives.
Sub-Clause 4.8 Health and Safety Obligations	Pursuant to clause 4.8(i), the health and safety manual shall set out all the health and safety requirements mentioned as follows: GUIDELINES FOR HEALTH AND SAFETY MANUAL The Contractor will prepare a health and safety manual by covering, but not limited to, the following aspects: The Contractor shall prepare, submit & implement the H&S Manual in accordance with the national & international requirements and as per Bank requirements / satisfaction, customized to local conditions as applicable. Review of Legal and Policy Regimes including labor laws, labor policy, occupational health and safety legislation and industrial legislation. Punjab Occupational Safety and Health Act, 2019 is the main legislation to be reviewed and the manual needs to abide by its clauses. Follow procedures and protocols for work health and safety and prepare procedures for workers' safety at the workplace including the provision of workplace safety, adequate staff training including topics such as safe work procedures, infection control procedures and appropriate hygiene, properly maintained facilities and equipment, including the provision of personal protective equipment such as gloves, eye protection and sharps containers; and a clean and suitably designed workplace with the safe storage of goods such as chemicals. The following procedures and standards would be made part of the manual. • maintain a safe, clean and efficient, working environment; • implement procedures and practices, in a variety of situations, in accordance with state and local government health regulations; • store and dispose of waste according to health regulations; • store, clean and disinfect linen in line with local health regulations; • clean walls, floor and working surfaces to meet health and safety standards without causing damage; • check all equipment for maintenance requirements; • refer equipment for repair as required; • store equipment safely; • identify fire hazards and take precautions to prevent fire; • safe lifting and carrying techniques maintained; • ensure student safety at all times; • ensure procedures for operator safety are followed at all times; • all unsafe situations recognized and reported;

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	• implement regular fire drills and provide first aid to all staff and students if needed; • display first aid and safety procedures for all staff and students to see; and • report any identified occupational health and safety hazard to the appropriate staff member as required. • All workers, Contractors, students & visitors must comply with safe work practices, with the intent of avoiding injury to themselves and others and damage to plant and equipment. The Contractor will prepare procedures for comprehensive hazard identification and risk assessments of all its operations including its premises. The Contractor will prepare a plan for risk management. All workers must wear appropriate PPE, including high-visibility clothing that is clearly visible to drivers, safety helmets, and safety shoes. The Contractor will prepare a critical incident management policy and make it part of the manual. This policy may include measures for physical safety. The policy needs to include measures for avoiding incidents and, in case of occurrence efficient management processes to address the incidents. The Contractor will prepare capacity-building plans for the training of staff on occupational health and safety and make part of the manual. The plan should include the content, frequency of training, follow-up training plan and reporting. The Contractor will prepare Standard Operating Procedures (SOPs) for worksite operations and make part of the manual. SOPs issued by WASA vide No. MD/WASA/653-55 dated 04.08.2025 on the directions of honorable High Court Lahore shall also be adhered, the copy of which is annexed in Appendix ER-04. The Contractor will prepare procedures for handling immediate emergency conditions such as fire, smoke, or other calamities. The Contractor shall prepare, submit, and implement the following plans (also reflected in Clause 5.2), customized to project/local conditions and to the satisfaction of the Employer and Bank: Design Phase Plans / Instruments (Prepared based on surveys, baseline studies, stakeholder consultations, and design inputs. To be finalized, approved, and later implemented during construction and operation.) • Risk Management Plan (hazard identification and risk assessment). • Critical Incident Management Policy. • Traffic and Road Safety Management Plan.
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	• Community Health and Safety Management Plan. • Labor Management Plan / Workers’ Camp Management Plan (framework prepared at design stage; detailed before mobilization). • GBV/SEA/SH Action Plan (Codes of Conduct, awareness, reporting procedures). • Grievance Redress Mechanism (GRM) Procedures. • Environmental Management Plan (EMP) (site-specific mitigation, monitoring, and reporting requirements). • Personnel safety & security plan Construction & Operation Phase Plans (Activated and implemented on the ground during execution and extended into operations where applicable.) • Standard Operating Procedures (SOPs) for all worksite operations. • Emergency Response and Preparedness Plan (fire, natural disasters, hazardous spills, calamities). • Waste Management Plan (solid, hazardous, wastewater handling, disposal, drainage, solar panel waste/PVC disposal/recycling). • Occupational Health Surveillance and Monitoring Plan (periodic health checks, medical records, prevention). • Infectious Disease Prevention & Control Plan (including COVID-19 or similar). Pursuant to clause 4.8, the Contractor shall maintain records and make reports as follows: The Contractor will prepare essential reporting formats and make part of the manual. The Contractor shall require its SubContractors and suppliers (other than SubContractors) to immediately notify the Contractor of any incidents or accidents referred to in this Sub-clause. Incident and Accident Reporting The Contractor is responsible for reporting immediately, but no later than 24 hours, any incidents, accidents, or allegations occurring during the execution of the contract as follows: 1. Reporting to the Employer In the event of an accident during the execution of the works, the Contractor shall immediately notify the Employer as per the conditions outlined in the contract. 2. Reporting to the Project Manager
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	The Contractor must inform the Project Manager immediately upon becoming aware of any: • Allegation, incident, or accident at the site that has or is likely to have a significant adverse effect on: The environment. • The affected communities or the public. • The Employer's Personnel or the Contractor's Personnel. • Incidents or accidents involving: - Fatalities or serious injuries. - Significant adverse effects or damage to public/private property. • Allegations of Gender-Based Violence (GBV), Sexual Exploitation and Abuse (SEA), or Sexual Harassment (SH). • For SEA or SH allegations, while maintaining confidentiality where appropriate, the Contractor shall include details such as: • Type of allegation (e.g., sexual exploitation, abuse, or harassment). • Gender and age of the affected individual. 3. Reporting on SubContractors or Suppliers The Contractor must also immediately notify the Project Manager of any incidents or accidents occurring on the premises of its SubContractors or suppliers that may have a significant adverse effect as described above. 4. Notification and Detailed Reporting • Initial notification must include sufficient detail about the incident or accident. • The Contractor shall provide a detailed report on the incident or accident to the Project Manager within a timeframe agreed upon with the Project Manager. Monitoring Reports • Daily ESHS Checklist Monitoring (covering site inspections, worker safety, use of PPE, housekeeping, waste management, and compliance observations). • Initial Notification Report (basic details of the incident/accident). • Detailed Incident/Accident Report within agreed timeframe. • Monthly ESHS Compliance Report (including health & safety statistics, trainings conducted, inspections, and grievances received). • Quarterly Environmental and Social Performance Report (to Employer/Bank as required). • Annual ESHS Summary Report (including lessons learned, corrective actions, improvements). • Carbon Emissions / GHG Monitoring Report (as part of sustainability reporting). Responsibility for SubContractors and Suppliers • The Contractor shall require its SubContractors and suppliers to immediately notify the Contractor of any incidents or accidents described in this sub-clause.
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	The Contractor is responsible for consolidating and reporting these incidents or accidents to the Project Manager.
Sub-Clause 4.9 Quality Management and Compliance Verification System	Pursuant to Sub-clause 4.9.1, the QMS shall be in accordance with following: The Contractor Quality Management System and Compliance Verification System shall meet all the requirements of the International Standards Organization known as ISO 9001. As appropriate the Contractor will impose on his design team, vendors, fabricators and sub- Contractors the requirement of ISO 9001.2015 Quality Management System, ISO 45001.2018 Occupational Health and Safety Management System, and OHSAS 18001:2007 Occupational Health and Safety. Pursuant to Sub-clause 4.9.1, the compliance verification system (CVS) shall be in accordance with following: The CVS should be a continuous cycle of planning, doing, checking, and acting, applied to every project element in the following manner: Define team & their responsibilities The Contractor shall depute QA/QC team whose responsibilities should be well defined. Identify standard and performance Parameters The Contractor shall Follow the standards (as mentioned in ER relevant sections i.e. Clause 5.2 & 5.4 of ER). For all components and activities, identify the exact contractual requirement from: - Employer's Requirements (Technical Specifications, Performance Criteria) - Applicable Laws and Regulations - Recognized International Standards (e.g., ISO, ASTM, EN, BS) - Manufacturer's recommendations for equipment. Plan the Verification - The Inspection & Test Plan (ITP) The Contractor shall Develop an ITP for each major deliverable, material, and construction activity as per clause 5.2, 12.1 of ER. The ITP should define all relevant details for testing & inspections (i.e. What, How, When, Hold Point, Witness Point, Surveillance Point, Accepted performance criteria in line with Clause 9.1 of ER) Execute the Work & Self-Verify The work crew or subContractor performs the activity according to the approved method statement. They conduct initial self-checks and record basic quality control data (e.g., setting out points, preliminary measurements). Perform Independent Verification & Record The Contractor's own Quality Assurance (QA) team (independent from the production team) performs inspections and tests as stipulated in the ITP. They complete checklists, test reports, and inspection forms. Seek Employer's Verification (If required by ITP) For Witness (W) and Hold (H) points, the Contractor shall formally notify the Employer within the agreed timeframe (e.g., 24-48 hours in advance).

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	The Employer will witness the inspection/test and signs the ITP or checklist to acknowledge it has been inspected. The inspection may be video/ photo recorded as required. Their signature is a record of attendance, not approval of the quality (which remains the Contractor's responsibility). • Address Non-Conformance In case of non-compliance, it will be immediately be reported to the Employer and accordingly a Non-Conformance Report (NCR) will be required to be issued. The NCR process involves: 1. Containment: Isolating the non-conforming product/work. 2. Root Cause Analysis: Investigating why it happened (e.g., 5 Whys, Fishbone Diagram). 3. Corrective Action: Fixing the specific issue (e.g., repair, replace, re-execute). 4. Preventive Action: Updating procedures/methods to prevent recurrence. 5. Follow-up Inspections: Follow-up inspections shall be carried out by the Contractor till desirable quality assurance • Release for Next Stage If Compliant, the ITP is signed off by the Contractor's QA, the work is released, and the next activity can begin. • Reporting Quality Management & Verification Compliance report & Compile As-Built Records All successfully completed checklists, test reports, and signed ITPs are compiled into the final quality records package. QMS Registry shall be maintained. These will be submitted in the form of report. Instantaneous NCR, Monthly reports, summary in Yearly reports. These become part of the As-Built documents and are handed over to the Employer, proving compliance.
Sub-Clause 4.16 Transport of Goods	Pursuant to Clause 4.16, the Contractor shall notify the Employer for any Goods that the Contractor intends to import into the Country—such as MTBM, Vertical Spinning Machines, Heavy Machinery i.e. crane, construction equipment, equipment forming part of the permanent works, materials, etc. The Contractor must first obtain the Employer's permission as per time specified in clause 4.16.
Sub-Clause 4.18 Protection of Environme nt	Pursuant to clause 4.18, the Contractor shall: Legal and Regulatory Compliance: • Adhere to the applicable clauses of the Punjab Environment Protection Act 2012 to ensure environmental protection. • Comply with the Punjab Environmental Quality Standards (PEQS), ensuring that no pollutants are discharged, emitted, or released into the environment beyond the limits specified in the PEQS. • Ensure that emissions, surface discharges, effluent, and any other pollutants resulting from their activities do not exceed the limits specified in the Contract Specifications, and the standards prescribed by applicable laws and regulations.

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	• Prevent nuisance to people and property caused by pollution, noise, or other impacts of the Contractor's operations or activities. • The Contractor shall also comply with all environmental and social management instruments including all plans as specified under Sub-Clause 4.8 of the Employer's Requirements. In case of any conflict during the the project between local guidelines or the Asian Infrastructure Investment Bank (AIIB) Environmental and Social Policy (ESP) requirements, the Contractor shall comply with the most stringent applicable standard. Environmental Protection Measures The Contractor shall take all necessary actions to: • Protect the environment, both on and off the Site. • Minimize damage and prevent nuisance to people and property caused by pollution, noise, or other impacts of the Contractor's operations or activities. Climate Change Adaptation and Mitigation The Contractor shall integrate climate change adaptation and mitigation measures into their operations and activities, as specified in the Contract and applicable laws. • The Contractor shall also quantify greenhouse gas (GHG) emissions from its operations (including fuel use, electricity consumption, transportation, and materials), applying internationally recognized methodologies such as the Intergovernmental Panel on Climate Change (IPCC) Guidelines (2006 and the 2019 Refinement), the International Organization for Standardization's ISO 14064 series (ISO 14064-1:2018, ISO 14064-2:2019, ISO 14064-3:2019), and the Greenhouse Gas Protocol. • In line with Pakistan's commitments under the Paris Agreement (2015), Pakistan's Nationally Determined Contribution (NDC), and applicable provincial climate policies, the Contractor shall support the Employer in identifying opportunities for emissions reduction, low-carbon construction methods, and potential carbon credit registration using UNFCCC-approved or voluntary carbon market methodologies. • Given the inclusion of a 450-kW solar power component, the Contractor shall ensure compliance with renewable energy regulations applicable in Pakistan, including those issued by the National Electric Power Regulatory Authority (NEPRA) and provincial authorities. The Contractor shall account for the emissions savings from renewable energy generation in the overall GHG emissions balance. • The Contractor shall provide verifiable GHG emissions data to the Employer as part of regular reporting, including baseline emissions, emissions reductions, and credits achieved, enabling the Employer to claim credit for verified emissions reductions or carbon credits. Pollutant Control
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	The Contractor shall ensure that emissions, surface discharges, effluent, and any other pollutants resulting from their activities do not exceed: • The limits specified in the Contract Specifications, and • The standards prescribed by applicable laws and regulations. Environmental Damage and Remediation If the Contractor's operations result in environmental damage, property damage, or nuisance to people on or off-site, the Contractor shall: • Coordinate with the Project Manager and Employer to agree on appropriate actions and a timeline to remediate the damaged environment, restoring it to its former condition as far as practicable. • Implement the agreed remedial actions at their own cost, ensuring satisfaction of the Project Manager and Employer. • The Contractor shall carry out all activities in a manner that does not cause harm to the environment, people, or property, either on or off the Site. In the event of such harm, the Contractor shall immediately stop the harmful activity, notify the Project Manager, implement remedial actions at its own cost, and indemnify and hold harmless the Employer against all claims, fines, penalties, and liabilities arising from such harm. Monitoring and Reporting • The environmental baseline monitoring carried out during the preliminary stage by the PMC through third party laboratory, as information, is attached in Appendix ER-03 of this document. • The Contractor shall ensure that project activities do not further deteriorate parameters already exceeding PEQS/WHO guidelines (including PM2.5, PM10, Suspended Particulate Matter, noise levels at Garhi Shahu and Shama Stop, and groundwater contamination at Shama Stop and Gulshan-e-Ravi). • The Contractor shall carry out environmental baseline monitoring of all parameters (parameters specified in ESMP) right after commencement of the project, through the EPA-certified laboratory meeting ISO 9001:2015 standards. • The ESMP is attached as Appendix-ER-03 of this document for implementation by the Contractor. • The locations of sample and methodology of sampling and testing shall be agreed with the Employer, before start of monitoring. Results shall be compared against PEQS and AIIB standards where applicable. • The Contractor will carry out baseline monitoring of all parameters during detailed design and construction phase, through EPA-certified laboratory meeting ISO 9001:2015 standards, as specified in ESMP.
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Sub-Clause 4.19 Temporary Utilities	Pursuant to the Clause 4.19, the Contractor shall be entitled to use, for the purposes of the Works, the utilities on the Site for which details and prices are given as follows: Water Source for Micro-Tunneling and construction activities: • It is recommended that the Contractor drill pumping wells to ensure an adequate water supply during the micro-tunneling process. • The location designated for the Contractor to drill pumping wells and extract water is shown in Figure LW-TD-06. • The Contractor must adhere to the guidelines and specifications outlined in this figure while planning and executing the drilling and water usage activities. • The location for such wells should be coordinated and agreed upon with relevant stakeholders to avoid any disruptions or conflicts. • Any deviations from the indicated location shall require prior approval from the Employer and WASA. • The Contractor shall ensure that extraction of groundwater does not adversely impact local communities, existing users, or the surrounding environment, in line with the Punjab Environmental Protection Act (2012) and AIB Environmental and Social Policy (2016). • The Contractor shall ensure that the quality of water used for micro-tunneling operations meets relevant technical standards (including sediment load, salinity, and chemical composition) to avoid damage to machinery and works. Drilling Wells at Construction Camps The Contractor may drill wells at construction camps, subject to prior permission from WASA. Compliance with Regulations • The Contractor must ensure compliance with all applicable WASA rules and regulations when utilizing water from hydrants or drilling wells. • Measures should be taken to minimize wastage and ensure sustainable use of water resources. Use of Electricity: The Contractor shall install electricity meters for construction, camps and offices after approval from LESCO as applicable under their Policy and Tariff. Use of Suigas: The Contractor may avail gas for cooking/ heating purpose through metered connection for their camps and offices after approval from SNGPL as applicable under their policy. LPG cylinders is an alternate facility that can be availed. Use of internet: The Contractor is also responsible for provision of internet facilities, at least from two different sources, having speed of at least 100 Mbs for offices and sites from best
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	available internet service providers. The contractor is responsible to include these provisions for design, construction and DNP period.
Sub-Clause 4.20 Progress Report	Pursuant to Sub-Clause 4.2 of the Conditions of Contract, the Contractor shall prepare and submit the progress reports covering the following aspects: CONTRACTOR'S REPORTING OBLIGATIONS The Contractor shall record video evidence of all key construction activities at all sites and obliged to submit with the Progress Reports. The Contractor shall submit the following reports during the currency of the Contract: • Daily Progress Report • Monthly Progress Report • Quarterly Progress Report • Half-Yearly Progress Report • Yearly Progress Report • Completion Report • Progress Report for Audit purpose • Progress Report for Inspection & Review meetings DESCRIPTION OF DAILY PROGRESS REPORTS A concise summary of daily progress reports at the beginning of the monthly report. Manpower Status • Details of international and national manpower working at the Site, including: ➤ Name, designation, national ID Card, Police Character certificate, contact details, visa & passport expiry status and role. • Identification of shortcomings in manpower arrangements, with reasons for any deficiencies. • Status of manpower replacements, if applicable, with justifications. Plant, Equipment & inventory Status • Utilization and status of required plant and equipment during the reporting month. • Identification of deficiencies or underutilized equipment, if any. • Pipes in stock & under construction v/s construction capacity • Deployment of machinery • Drive length achieved for each active site • Reasons of inactivity against each site • Total Operating Hours of MTBM Machine • Total Progress achieved at each site • Status of sheet piling

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	• Status of excavation • Status of shifting of services • Bottlenecks requiring Employers attention • Details of alerts notified to site management, PMC & Employer Construction Material Status Status of required construction materials, including: • Utilization, surplus, and shortages. • Reasons for shortages and proposed arrangements for procurement. Contractual Matters Updates on: • Variations (Sub-Clause 13). • Claims (Sub-Clause 20.2.1). • Arbitration cases (Sub-Clause 21). Dispute Avoidance/Adjudication Board (DAAB) Activities • Summary of DAAB meetings, conclusions, and recommendations. • Agreements reached as a result of DAAB recommendations. Exceptional Events • Description of any exceptional events during the reporting month or year. • Proposed way forward to manage impacts. Insurances and Guarantees • Status of insurances and guarantees as per the Contract and applicable country laws and regulations. • Identification of any gaps and required actions. Financial Progress Comprehensive financial updates, including: • Status of payment of invoices, withheld amounts, and re-invoicing. • Cash flow requirements for the next month, quarter, six months, and year. • Comparison of cash flow demand versus expenditure and identification of any surplus/deficit. • Claims arising from delayed payments, if any.
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	<i>Employer/PMU & PMC Performance</i> • Assessment of assistance, reviews, and approvals provided by the Employer, PMU, and PMC. • Identification of delays or issues in their performance. • PMC staff supervision personal name, designation, and on-site time. DESCRIPTION OF THE MONTHLY PROGRESS REPORT The Contractor shall prepare and submit the Monthly Progress Report as per the requirements outlined in Sub-Clause 4.20 of the GCC. Additionally, the report must include the following details: The Contractor shall submit a comprehensive Monthly Progress Report which shall contain all details which are linked with the payment schedule in PART-III Section 9 (Contract Forms) in Particular and all other important updates in general. It shall be submitted to the Employer's Representative (and Project Management Consultant - PMC) by the 5th day of each month, covering the period from the first to the last day of the previous month. The report shall provide a detailed, accurate, and verifiable account of the progress achieved against the approved Programme, the resources deployed, and all critical issues affecting the Project. The primary purpose of this report is to enable the Employer to cross-verify the progress claimed in the Contractor's monthly Statement (Interim Payment Application) against physical, on-ground achievements and documented deliverables. 4.2.2 Content of Monthly Progress Report The report shall be structured to include, but not be limited to, the following sections: A. Executive Summary: Overall project status (Ahead, On, Behind Schedule) against the approved baseline Programme. Summary of progress during the reporting period for key project components (Design, Procurement, Construction-Part 1, Construction-Part 2). Key milestones achieved in the reporting period. Key issues, risks, and mitigation plans. Health, Safety, and Environmental (HSE) performance summary. B. Detailed Design Progress (Measured by Deliverable Submissions & Approvals): The report shall include a table with the status of each design deliverable. This table shall be the basis for approving payment for the design phase with atleast following
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	- Graphical S-Curves showing Planned Value, Earned Value (BCWP), and Actual Cost (ACWP) for the project overall and for key components. - Critical Path Analysis and explanation of any delays. F. Quality, HSE, and Documentation: Summary of ITPs raised, inspections held, and non-conformances reported. HSE statistics (man-hours, incidents, near-misses, trainings). List of all drawings, method statements, and materials submittals submitted and approved during the period. 4.2.3 Link to Payment Certification The Employer's Representative/PMC will use this Progress Report to verify the quantities and milestones claimed in the Contractor's monthly payment application. Payment for construction work will be based on the measured quantities and achieved milestones as verified and reported in this document. Discrepancies between the payment application and the verified progress report will be queried and must be resolved before the payment certificate is issued. 4.2.4 Supporting Evidence - The report shall be supported by: - Photographs: Date-stamped, geotagged photographs of all reported activities and progress. - Survey Reports: For pipe laying grades, alignments, and finished levels. - Inspection Records: Signed copies of all ITPs and quality control records. - Delivery Notes: For all materials and equipment delivered to site. QUARTERLY, HALF-YEARLY, AND YEARLY PROGRESS REPORTS - These reports shall summarize the details of Monthly Progress Reports and include updates for the current month. - The focus will be on cumulative progress, performance evaluation, and projections for the upcoming period.
Sub-Clause 5.1 General Design Obligations	Pursuant to Sub-Clause 5.1(b) (Contractors' documents) of the Conditions of Contract, the eligibility criteria for Contractors' Design Team is as follows: Pursuant to Sub-Clause 5.1(b) of the conditions of the contract, the eligibility criteria of Contractors design team is same as mentioned in Section-3 "Evaluation & Qualifications Criteria" Pursuant to Sub-Clause 5.1 (Contractors' documents) of the Conditions of Contract, the Design Criteria is as follows:

DESIGN CRITERIA

Pursuant to Sub-Clause 5.1 of the conditions of the contract, the Design Criteria of various project components including Civil, Mechanical, Electrical, SCADA etc is mentioned hereunder:

PART-I

CIVIL COMPONENTS

The following are the main structures and equipment shall be constructed and installed. However, the EPC Contractor have the right to increase or decrease this scope after the approval of the Employer;

- Construction of 23 No. Intake Structures Consist of Inlet Chamber, Screen Chamber, proportional weir, Outlet Chamber, Connection system of inlet chamber to existing sewer. These structures shall preferably be placed in the premises of existing disposal station to be eliminated after trunk sewer in operation.
- Construction of Bypass arrangement at Existing lift stations through laying of bypass sewers that will be laid inside lift stations to ensure uninterrupted operation of those pumping station during monsoon or any untoward situations.
- Construction of 17 No. Drop Structures Consist of Inlet/ Approach Channel, Drop Shaft, Deaeration Chamber, Adit, etc.
- Connection with existing sewerage system including new manholes and RCC sewer pipes, excavation, laying backfilling, bypass arrangement dismantling and reconstruction of existing structure where required.
- Construction of 31 No. Odour Control and Ventilation system consist of fans, activated carbon filters, etc. in room with different size as per sewer intake flows.
- Connection Tunnels of 23 Nos. between Adits and nearest manholes at each disposal stations.
- Manufacturing of reinforced concrete pipes with 6000 PSI strength or higher, with HDPE lining, of different sizes and length such as;

Sr. No	Inner Diameter (m)	Length (m)
1	3.5	5,337
2	2.4	11,582
3	1.5	5,454
4	1.05	9,837

- Laying of above-mentioned concrete pipes with the following size of micro-tunnelling machines

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Sr. No	MTBM Diameter (m)	No. of MTBM
1	3.5	1.0
2	2.4	2.0
3	1.5	1.0
4	1.05	2.0

• The following was assumed during preliminary designing and the Contractor may use or can change the size

#	Pipe Inner Diameter (mm)	Pipe Outer Diameter (mm)	Drive Length (m)	Cutter head Diameter (mm)	Space for Cement Slurry Grouting (mm)
1	1,050	1,350	190	1,375	25
2	1,500	1,838	250	1,875	38
3	2,400	2,850	350	2,900	50
4	3,500	4,100	500	4,150	50

• The following was worked out during preliminary designing:

Diameter (mm)	Lengths (m)	No. of Jacking Shafts	No. of Receiving Shafts
1050	9837	32	30
1500	5454	13	16
2400	11582	27	26
3500	5337	10	11
Total	32210	82	83

- Construction of 165 Nos. of Jacking and Receiving Shafts with Steel Sheet Piles Internal Braced with Steel Elements with different depths.
- Construction of 91 Manholes of reinforced concretes
- Construction of Ventilations and Odour Control System at selected Manholes
- Installation of SCADA system and Intake and along the lines and laterals at selected Manholes.
- Rehabilitation of Roads at Jacking and Receiving Shaft after backfilling.
- Rehabilitation of Cantonment Drain by lining
- Construction of rider sewer at required location.
- Laying of Jhanki Devi Sewer with Open Cut having length of 159 m.
- Construction of Generator foundation and shifting at new location.
- Construction of control rooms at 23 No. location of decommissioned disposal stations.

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	• Construction of rainwater intake and connection with trunk sewer through drop manholes • Construction of Drop Manholes at selected location • Construction of drop structures along Line-A and Line-B near Qartaba Chowk. PART-II Construction of a disposal station equipped with: • 16 pumps, each with a discharge capacity of 1.5 m³/s, • A concrete channel, • A distribution chamber, • A concrete conduit, • Transition structure for connection of proposed conduits and existing outfall drain • A head regulator/Penstock gate, • Rehabilitation of the outfall disposal drain using brick lining along both sides up to the River Ravi. • Installation of SCADA Control Room at disposal station. • Installation of Solar system • Electrical grid station/GIS grid or Connection with existing Grid station through a dedicated feeder HYDRAULIC DESIGN CRITERIA ○ <u>Concrete Pipes</u> <i>Pipe Material and Design Standards:</i> • Reinforced Concrete Pipes (RCP) are recommended for pipe jacking operations based on the proposed diameters and locally available materials. • The design shall adhere to ASCE 27-17 guidelines and conform to ASTM C-76-20, and WASA design criteria accounting for temporary loads during installation (e.g., lifting, transportation, annular grouting, and jacking) and permanent loads (e.g., earth load, hydrostatic pressure, and surcharge load). • Design of Pipe/sewer should be based on flows assessment in accordance with Population forecasted on Geometrical Growth rate method and Population census of 2017 published by Pakistan Bureau of statistics (same as already established in PMC Prepared PEDR) <i>Sewerage System Design Capacity:</i>
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Flow rate of each point source are given below.

Route of Sewer	Contributing Area	Flow (ft ³ /sec)
Karachi Phatak to Infantry Road / Allama Iqbal Road crossing	Harbanspura + Al Faisal Town + Jorry Pull + Aziz Bhatti Town + Haji Pura + Jalal Colony + Sally Town + Taj Pura + Manzoor Colony + Taj Bagh Scheme + Saddar DS+ Ghosia Phatak DS+ Shalimar DS +Askari-IX DS+ Sofia Trust DS	170.12
Mian Mir DS to Infantry Road	Mian Mir + Upper Mall	15.39
Infantry Road DS to Infantry Road / Allama Iqbal Road crossing	Infantry	69.25
Infantry Road / Allama Iqbal Road crossing to Bird Market D/S		230.04
Bird Market DS to Mughalpura Road / Allama Iqbal Road junction	Bird Market	251.56
Larech Colony to Garhi Shahu Bazar	Larech Colony	2.08
Ghazi Mohallah DS to Garhi Shahu Bazar	Ghazi Mohallah	2.35
Garhi Shahu Bazar to Mughalpura Road / Allama Iqbal Road junction		4.43
Mughalpura / Allama Iqbal Road Junction to the start of Durand Road		254.0
Muhammad Nagar to the start of Durand Road	Muhammad Nagar	1.4
Durand Road / Allama Iqbal Road junction to Davis / Durand Road Junction		255
Governor House to Davis Road/ Durand Road junction	Governor House	9.7
Davis Road / Durand Road Junction to start of Edgerton Road Shimla Hill		261.5
Lakshmi Chowk DS to Edgerton Road Shimla Hill	Lakshmi Chowk	83.03
Edgerton Road Shimla Hill to Qartaba Chowk		328.83
Ganga Ram Hospital Emergency Chowk to Qartaba Chowk		328.83
Gurumangat DS to College Road / Q Block Llink Rd Crossing	Gurumangat + Khan Colony + Sultan Park	59.08

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	Q Block Gulberg DS to College Road / Q Block link Rd Crossing	Q Block Gulberg + Fatehabad	32.05
	College Road / Q Block link Rd Crossing to Gulberg Main Boulevard		82.01
	Central Point DS to Gulberg Drain along Main Boulevard		24.81
	Gulberg Main Boulevard / M. A. Gurmani Road Crossing to G Block Gulberg DS		103.27
	Mian Mir DS to Zafar Ali Road DS	Mian Mir+ Abdul Rehman Road DS+ CMH Area	39.79
	G Block Gulberg DS to Canal Crossing	G Block Gulberg	134.09
	Zafar Ali Road DS to Shareef Colony	Zafar Ali Rd	4.09
	Shareef Colony to Canal Crossing	Shareef Colony	5.78
	Canal Crossing to Ichra/Ferozepur Road Crossing		137.47
	Ichra DS to Shama/Ferozpur Road intersection junction	Ichra	149.48
	Shadman DS to Shama DS	Shadman+ Mental Hospital+ PIC+ GOR III+ 25% of Jail Road	43.71
	Shama DS to Shama/Ferozepur Road Intersection		55.82
	Shama/Ferozpur Road intersection to Link Ferozepur Road Intersection		197.79
	Rasool Park DS to Link Ferozepur Road Intersection	Rasool Park	94.75
	Link Ferozepur Road intersection to Qartaba Chowk		286.61
	Qartaba Chowk to Chauburji Chowk		606.52
	Lytton Road DS to Chauburji Chowk	Lytton Road + Lake Road + High Court/Mozang	29.88
	PIA Planetarium to Near Chauburji chowk	PIA Planetarium	20.97
	PIA Planetarium to Chauburj Park Road	PIA Planetarium	47.22
	Chauburji chowk/Multan Road to Samanabd Morr		646.12

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	Poonch Road DS to Samanabad Morr	Poonch Road+ Bahawalpur Road+ Chauburji Quarters	75.08
	Samnabad Morr to Gulshan-e-Ravi DS		688.10 (cumulative)
• Main trunk sewer pipes are designed to handle peak flows estimated for the year 2050. • Pipes are sized to operate at 80% of their full flow depth to ensure proper ventilation and prevent septicity. At this depth: • $V/V_{full} = 1$ • $Q/Q_{full} = 0.88$ • The design flow is calculated by dividing Q by 0.88 to achieve the 80% flowing depth.			
Pipe Fabrication: • Pipes will be locally manufactured, requiring the EPC Contractor to establish a pipe fabrication unit at suitable location. • Pipes shall be 2.44 m (8 ft.) in length, with a concrete strength of 6,000 psi or higher, and doubly reinforced with spiral stirrups. However, EPC contractor may opt for different pipe length suiting design requirements. • The EPC Contractor shall maintain quality control, importing pipe testing rigs as per ASTM standards.			
Velocity in Pipes • Self-cleansing velocities are maintained between 0.9 m/s and 2.0 m/s to prevent sediment deposition. • A velocity of 0.9 m/s is used for design simplicity. • Velocity at low flows (present year during nighttime) shall not be less than 0.3m/s and preferably 0.6m/s. The Contractor must simulate the design on approved sewer design software and conduct EPS, extended period simulation and ensure calibration of software before performing simulations. Minimum flow factor must be determined by the Contractor through physical measurement.			
Coefficient of Roughness • Manning's coefficient of internal friction: • Unlined Concrete: 0.013 - 0.014. • HDPE Lined: 0.009 - 0.011. • High-density polyethylene (HDPE) lining is recommended to protect concrete from H_2S gas. • All sewer pipes shall be HDPE lined, fixed during the construction of the HDPE pipe.			

- Any structure where HDPE lining is not possible, GRP lining or Epoxy coating shall be applied as approved by the Engineer.
- HDPE Lining of Pipes: 3 mm thick (T-Grip Liner using Virgin HDPE material)

Gradient of Sewer Pipes

Sewer gradients align with WASA Design Criteria and adjust to accommodate natural topography and permissible velocity limits.

The minimum gradient of sewers would generally be as per design requirement to fulfil the velocity criteria as shown in the following table:

Recommended Gradients for Various Pipe Sizes

Pipe Internal Diameter (mm)	Slopes
1050	0.0006-0.0029
1500	0.0004-0.0019
2400	0.0007-0.0017
3500	0.0006 – 0.0009

○ **Design of Storm Water Inlet**

Types of Inlets:

- Type A: In sumps with curb opening, grate, or combination (grate & curb opening).
- Type B: On grade without gutter depression (curb opening, grate, or combination).
- Type C: on grade with gutter depression: On grade with curb opening, grate, or combination.

Design Approach:

- Combination-type inlets are generally used except at Lakshmi Chowk, where only grate types are employed.
- Gutter flow depth must not exceed permissible limits during the design runoff event.

The allowable depth of gutter flow is a function of the height of curb and is given in the table below.

Table 4.12: Allowable Depth of Gutter Flow

Allowable Depth of Gutter Flow	
Height of Curb	Maximum Depth of Flow

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(in.)	(mm)	(ft.)	(mm)
3	75	0.200	60
4	100	0.267	80
6	150	0.400	120
8	200	0.450	135

In no case should the depth of gutter flow exceed the values given in the above table during the design runoff event.

Sizing of Manholes

- Manholes are required at all changes in direction or alignment, changes in sewer size, and at junctions where sewers of different sizes connect.
- Bottom-level sizing depends on the pipe's outer diameter plus 1.0 m (0.5 m on each side for benching).

Sizing of Jacking and Receiving Shafts

The following size were assumed during preliminary design phase.

#	Pipe Inner Diameter (mm)	Drive Lengths (m)	Jacking Shaft Dimensions			Jacking Shaft Dimensions		
			Rectangle		Circular	Rectangle		Circular
1.	1,050	190	4.5	7.6	9.0	2.5	3.5	5.0
2.	1,500	250	5.0	8.2	10.0	2.5	4.5	6.0
3.	2,400	350	6.0	8.8	11.0	3.5	5.5	7.0
4.	3,500	500	7.0	8.8	12.0	4.0	6.5	8.0

Sizing of Thrust Block

Two thrust blocks of reinforced concrete will be constructed in each Jacking shaft with the dimension given in below Table.

#	Pipe Inner Diameter (mm)	Pipe Outer Diameter (mm)	Cutter head Diameter (mm)	Size of Thrust Block	
				Length (m)	Width (m)
1	1,050	1,350	1,375	2.5	2.5
2	1,500	1,838	1,875	3	3
3	2,400	2,850	2,900	4	4
4	3,500	4,100	4,150	5	5

STRUCTURE DESIGN CRITERIA

Stability Criteria

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Stability analysis will be carried out for structures for most severe conditions of horizontal and vertical forces. Stability criteria for the gravity structures are established for three general loading conditions as follows:

Normal Loads: Normal loads are those which have a reasonable probability of occurring simultaneously, and where the various canals, river, tail water, and ground water levels are likely to persist for extended periods of time.

Exceptional Loads: Exceptional loads are those loads in which the temporary effects of earthquake forces are combined with normal loads or high flood condition is combined with normal loads.

Extreme Loads: Extreme loads are those combinations which could occur simultaneously but whose simultaneous occurrence would be expected only rarely, and for short duration.

Loading Conditions

The following cases will comprise the loading combinations:

Case 1	Construction	Dead load only
Case 2	Normal operation	Normal headwater + highest tail water + service loads.
Case 3	Inspection & Maintenance	One regulator bay empty, otherwise as in Case 2
Case 4	Emergency	Full headwater, no tail water
Case 5	Earthquake	As in case 2 + seismic loads.

Safety Factors

Stability safety factors for the structures will not be less than the values shown in the below table.

Loading Case	Overturning	Sliding
Case 1	1.50	1.50
Case 2	2.00	1.50
Case 3	1.75	1.50

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	Case 4	1.75	1.50
	Case 5	1.25	1.30

Material Strengths

The following grades of concrete and reinforcement is given in table below will be considered in designing concrete structures.

Type of structure	6-inch dia. Cylinder strength at 28 days	Steel reinforcement
Disposal Station (Dry & Wet well)	27.6 MPa (4000 Psi)	Grade 60 ASTM A615
Intake structure	20.7 MPa (3000 Psi)	Grade 60 ASTM A615
Stilling basin appurtenant	27.6 MPa (4000 Psi)	Grade 60 ASTM A615
Concrete piles	27.6 MPa (4000 Psi)	Grade 60 ASTM A615
Road and regulation bridges	27.6 MPa (4000 Psi)	Grade 60 ASTM A615
RCC Sewer Pipe	41.4 MPa (6000 Psi)	Grade 60 ASTM A615

- All reinforcement steel will be deformed bars of Grade 60 and 40 having minimum yield strengths of 60,000 Psi and 40,000 Psi respectively as per ASTM A615.
- For gates, stop logs, hoist and other structural items including superstructures, the structural mild steel designated ASTM A36, ASTM A-242 or equivalent, will be used.
- The bricks will have a minimum crushing strength of 2,000 Psi, when tested flat.

Aggregates

The concrete aggregates will be of a quality as provided in the USBR "Concrete Manual" with the flowing maximum sizes:

- Mass concrete 3 in
- Plain and reinforced cast-in-situ concrete 1.5 in

- Precast members 1.0 in

Backfilling of each Site

The Contractor shall backfill the area, wherever applicable, with granular non-cohesive soil i.e. sand, as per Geotech investigations (as approved by the Employer) and compact it in layers. The backfilling shall be carried out in layers, each with a thickness of 150 mm to make sure the proper compaction and stability. The site shall be compacted by mechanical means at optimum moisture content and dressing to designed section complete in all respects with 95% to 100% maximum modified AASHTO dry density.

Design Loadings

Dead Loads

The dead loads consist of self-weight of the structure and permanent superimposed loads. The unit weights of various materials for computing dead loads are given in table below:

Material Type	Unit Weight	
	(KN/m ³)	(lb/ft ³)
Reinforced Concrete	24.0	150
Plain Concrete	23.0	144
Stone masonry	23.0	140
Water	9.8	62.4
Brick masonry	19.0	120
Stone masonry	23.0	140
Water	9.8	62.4
Dry earth	16.0	100
Compacted earth	18.0	120
Saturated earth	21.0	135
Steel	77.0	490
Bitumen	13.7	87
Cement	14.1	90

Live Loads

The live loads are for roofs, beams and floors etc. and for the structural elements of staff residential buildings. The unit loads of various components of a building for computing live loads are given in below table.

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Buildings	(KN/m²)	(lb/ft²)
Floor (Residential)	1.90	40
Roof (Residential)	0.95	20
Floor (Office)	3.80	80
Roof (Office)	1.90	40
Varandah, Balcony	3.80	80
Stairs, landings	4.80	100

Wind Loads

Wind pressure will be applied to the exposed area of all structures in accordance with Building Code of Pakistan, for a maximum wind velocity of 100 mph (161 kmph) acting horizontally in any direction.

Earthquake Load

The earthquake loading for project will be selected according to Seismic Hazard Analysis.

Water Pressure

Water retaining structure, support the water pressure from upstream and downstream also. It is minimum at top while maximum near foundation level.

Uplift Pressure

Uplift pressure will be assumed to correspond to full head across. Uplift will be assumed to act on hundred (100) percent of the base area. Stability of pump house and head-regulator raft against floatation will be determined without taking into account weight of pumps and gates and water.

Earth Pressure

Lateral earth pressures due to backfill under static conditions will be computed by the Coulomb's Method, taking into account the effects of any soil saturation or submergence. A surcharge of two hundred 975 kg/m² (200 lb/sft) will be added for computing earth pressure.

Super imposed load

Manholes, intake structures and jacking & receiving shaft shall be constructed by considering in addition to backfill load, a super imposed load of crane or heavy load vehicle for stability of walls.

Factored Load of Full Load Combinations

These are indicated as given below:

Condition	Factored Load or Load Effect U
Basic	$U = 1.40 + 1.7L$
Winds	$U = 0.75 (1.40 + 1.7L + 1.7W)$ and include consideration of $L = 0$ $U = 1.050 + 1.3W$
Earthquake	$U = 0.75 (1.40 + 1.7L + 1.87E)$ and include consideration of $L = 0$ $U = 1.050 + 1.40 E$
Earth Pressure	$U = 1.40 + 1.7L + 1.7H$ $U = 1.050 + 1.7H$
Fluids-normal max. water level	$U = 1.40 + 1.7L + 1.4F$ $U = 0.90 + 1.4F$
Fluids-extreme water flow (overflow)	$U = 1.40 + 1.7L + 1.25F$
Impact	Substitute $(L + 1)$ for L
Settlement, creep, shrinkage or temperature change effects	$U = 0.75 (1.40 + 1.4T + 1.7L)$ $U = 1.4 (0 + T)$

MANHOLE COVER SPECIFICATIONS

Covers for inspection manholes and frames shall be of Ductile Iron or composite material and rated as heavy duty. They shall comply with BS EN 124 (E600) for areas with very heavy traffic. The manholes shall be provided with Reinforced Concrete Cover slab. Covers shall be securely lockable and complete with lifting keys. The clear opening shall be as per respective drawings but minimum of 600 mm.

DESIGN OF HYDRAULIC STEEL STRUCTURE

GENERAL

The project include hydraulic steel structure equipment:

- Steel Gates at Head regulator/ Penstock of Outfall Channel
- Steel Gates at Intake of Concrete Conduits
- Steel Gates at Spill Section of Spill Channel
- Steel Gates at Concrete Channel of New Disposal Station

	• Steel Gates at Concrete Channel of Old Disposal Station <u>DESIGN CRITERIA</u> The design, fabrication, and installation of all gates, stop logs, and related hydraulic steel structures will be carried out in accordance with recognized international standards to ensure structural integrity, durability, and operational efficiency. <u>Applicable Design Standards</u> All hydraulic steel structures and their associated drive mechanisms will be designed and calculated as per the following international standards: • DIN 19704 – Criteria for design and structural analysis of hydraulic steel structures. • DIN 19705 – Guidelines for the design, construction, and installation of hydraulic steel structures. • Equivalent British Standards (BS) or American Standards (ASTM, AISC, AWWA, ASCE, ANSI, API) may be used where applicable. <u>Material Specifications & Structural Design Considerations</u> <u>Structural Steel Material</u> • Gates will be fabricated using structural steel, conforming to DIN EN 10025. • This ensures high strength, impact resistance, and durability under hydraulic loads. <u>Corrosion and Wear Protection</u> • Components exposed to water and not made of stainless steel will be coated with a wear-resistant corrosion protection system. • The system will comply with ISO 12944 (Corrosion Protection of Steel Structures) and ISO 4628 (Evaluation of Degradation of Coatings). • Special attention will be given to abrasive material transport and sediment loading, ensuring prolonged service life and minimal maintenance. <u>Sediment & Bed Load Considerations</u> • The expected rate of sediment transport and impact of bed load in the drain will be factored into the design.
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	• Structural reinforcements and protective coatings will be employed to mitigate wear and abrasion effects. By following these standards and material requirements, the gates, stop logs, and associated hydraulic steel structures will be engineered for optimal performance, longevity, and reliability in extreme hydraulic conditions. DESIGN CRITERIA FOR ELIMINATED LIFT STATIONS, INTAKE STRUCTURES DROP STRUCTURES, VENTILATION, AND ODOR CONTROL SYSTEMS This section outlines the design criteria and technical requirements needed for the hydraulic and mechanical design of sewage intake points situated at various locations, including the decommissioned lift stations, drop structures, and their associated hydraulic components. All design efforts must adhere to the specified design criteria, the Employer's requirements, mechanical technical specifications, and drawings. A new mechanical bar rack coarse screen shall be installed at each decommissioned lift station, while the existing assembly will remain operational as a standby system. After passing through the mechanical screen, sewage will be transferred to the trunk sewer via a drop structure (where applicable) through gravity flow. For each drop structure, an air extraction and odor control system shall be installed to treat harmful and odor-causing gases. To prevent the accumulation of unwanted gases and ensure proper ventilation in the trunk sewer, a sewerage ventilation system shall also be integrated into the trunk sewer design. This system must include air entry points, a fan station, and an air treatment system. • DESIGN OF INTAKE STRUCTURES General Requirements: • Existing sewage intake structures at the 23 disposal stations will remain intact, which includes existing screening system, wet well, pumping station, piping works and electrical systems. • Proposed intake structures, along with drop and ventilation structures, will be constructed within the premises of the existing stations to avoid land acquisition. • Where space is limited specific construction technique shall be adopted to reduce the space requirement for excavation in case of deep structures and also to protect the existing structure. • The EPC Contractor can demolish and rebuild any existing structure to create space for the construction of new components however it is the responsibility of the Contractor to rebuild all demolished structure within the available land considering furnished multi-storey buildings as an option.
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	• Additionally, all existing wet wells and other structures must be covered to prevent odour issues. Furthermore, the existing well should be rehabilitated where necessary. Components: • Inlet Chamber • Screen Chamber (automatic screens included, if sewage effluent allows) • Outlet Chamber • Drop structure • Odor control system Bypass Arrangement: Included for emergency situations during heavy rains or maintenance requiring shutdown of proposed intake structures. • INLET CHAMBER The purpose of the sewage Inlet chamber is to collect sewage from the existing sewerage system and transfer it to the new screening system. A bypass arrangement is also provided to divert the sewage towards the existing manual screen and wet well in case the mechanical screen malfunctions or emergency situations or during excessive flows in case of monsoon season. DESIGN CRITERIA The inlet chamber design shall prioritize the efficient and safe transport of sewage to the screening chamber or bypass system, while reducing solids accumulation and ensuring a seamless transition to the screening chamber, thereby preventing excessive turbulence and odor generation. Primarily, the dimensions of the inlet chamber should be determined by the sizes of the connected inlet and bypass pipes and outlet channels, ensuring adequate space for the installation of penstock gate valves and facilitating future operation and maintenance. The design must accommodate anticipated peak flow rates, including both average daily flows and peak storm conditions. Additionally, it should aim to minimize headloss within the diversion structure to avert backflow and promote efficient functionality. Consideration should also be given to the design of a bypass system to manage excess flow during extreme events or when the mechanical screening system requires maintenance or repair. The contractor shall provide, wherever the space permits at the lift stations, the following configuration that will consist of: • One active screen that continuously removes debris from the flow. • One standby screen that is available to automatically take over during the following scenarios:
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1. When the active screen is taken out of service for routine maintenance or repairs.
2. If the active screen becomes clogged or fails.

During periods of high flow, where both screens might be operated simultaneously to handle the increased load.

Design Criteria of Inlet Chamber

Description	Value	Units
Material of Construction	RCC	
Minimum effective detention time	6	seconds

• PENSTOCK GATE

Penstock gates, often referred to as Sluice gates, slide gates, etc., are sturdy constructions made from stainless steel to control or block the flow of water in channels or pipelines, guaranteeing efficient management of operation and maintenance activities. They are essential in directing water to various parts of a system.

In this project, penstock gates shall be used to isolate or divert the sewage flow rate to the screening chambers or bypass the system in case of any emergency or maintenance work. These gates allow for the diversion of sewage to the operational mechanical screen during normal operation.

DESIGN CRITERIA

The design criteria for sluice gate valves shall prioritize reliable and efficient control of water flow, strong structural integrity, and effective leak-proof sealing. The thickness of the gate and the configuration of beam frameworks shall be adequate to endure the water pressure and operational forces under double-acting conditions. The mechanism, such as the stem and handwheel, must possess sufficient strength to operate the gate against the expected hydraulic load and friction. The design of the water stop pressure plate is vital for ensuring a secure seal against water flow and must be robust enough to handle pressures while maintaining a tight seal at a minimum water head of 6 m. Adhering to standardized design practices and established guidelines (e.g., AWWA/BS/DIN standards) is essential for guaranteeing quality and safety. The valve should be tailored to accommodate the specific hydraulic characteristics of the application, including maximum flow rates, pressure drops, and water depths. Additionally, the design should facilitate straightforward inspection, maintenance, and repair of the valves.

Design Criteria of Penstock Gate Valves

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Parameter	Design Criteria
Type of Penstock Gate	Channel Type or wall-mounted pedestal Type (in case of channel type sluice gate, a recess shall be formed in the RCC channel to avoid reducing the area of water flow)
Size	According to Dimensions of the mechanical screen, Bypass pipe sizes
Sealing System	3-side sealing for open channel flow 4-side sealing for isolation of pipe flow
Material of Construction	Main plate and frame construction: Stainless Steel SS316 Sealing: EPDM
Sealing design	Double acting
Design head	6 m
Operation	Manual, rising stem
Leakage Rate	Maximum allowable leakage according to BS 7775 for on- and off-seating heads up to and including 6 meters – 0.5 liters per minute per meter of seal perimeter

• MECHANICAL RAKED BAR SCREEN

The EPC Contractor shall comply with the following design criteria and requirements for the design of mechanical bar-racked screens:

DESIGN CRITERIA

The mechanical bar rake coarse screens shall be provided with design features incorporating chain-driven (multi rake).

If another type of bar screens (cleaning mechanism: reciprocating rake (with chain or hydraulic assembly), cable operated type for deep channels) are proposed to reduce the operation and maintenance cost, the Contractor must demonstrate in their Bid, through documentation and specifications, that the suggested equipment is proven and meets or exceeds the design quality, and performance requirements outlined in the design criteria.

The Client shall decide the most appropriate type of screening mechanism based on local conditions, cost, quality, reliability, proven design and low operation and maintenance cost.

The design criteria for the mechanical bar rack coarse screen are provided in the table below.

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Design Criteria for Mechanical Bar Screens			
Parameter	Unit	Range	Selected Value
Bar Size			
Width	mm	10~15	15
Depth	mm	30~38	30
Clear Spacing between Bars	mm	25-50	50
Slope from vertical	°	0~30	30
Approach Velocity			
Maximum	m/s	0.6~1.0	0.6~1.0
Minimum	m/s	0.3~0.5	0.3
Allowable Head loss	mm	150~600	250
Maximum velocity through the screen	m/s	0.9	0.9
The mechanical bar rack screens shall adhere to the design specifications and drawings, with one or a maximum of 2 screens based on available space at different lift stations, with each screen capable of handling the peak flow calculated. The stainless-steel SS316 bars used in the manufacturing of the screen shall have a minimum thickness of 15mm and a depth of 30~38mm. The Contractor shall ensure clear bar spacing of 50mm between the bars for efficient waste retention. The screen bar angle from the vertical should be in the range as specified 30° (70° from the horizontal). The depth of the screen channel and the total length of the screen shall be calculated as per the invert level of the trunk sewer. The Contractor is responsible for conducting a detailed topographical survey at each disposal station and verifying the invert levels and sizes of existing sewer pipes, screens, and wet wells. The Contractor must ensure the velocity of water passing through the screen bars does not exceed 0.9m/s to limit the head loss through the screen. The total screen chamber width shall ensure the minimum approach velocity, preferably at all flows, to avoid the settlement of solids. A proportional weir shall be designed and installed after each screen to maintain the velocity in the channel during low flow conditions to avoid the settlement of solids at the bottom of the screen channel.			
• DESIGN CRITERIA OF PROPORTIONAL WEIR			
Design Criteria for Proportional Weir			
Parameter	Formulas		Units

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Flow through a proportional weir	$Q = 1.57 \times Cd \times (2g)^{0.5} \times LH^{3/2}$	m ³ /s
Length	$L = Q / 4.173 (H)^{3/2}$	m
Cd	0.6	
Other configurations, if proposed by the EPC Contractor, must be technically justified with literature, design calculations, and manufacturer recommendations in their Bid. The complete submittal with technical literature and shop drawings shall be submitted by the Contractor for approval from the Engineer. The screen inlet chamber shall be equipped with a manual gear-operated penstock gate valve, for isolation of the screen chambers. The penstocks can also be used to bypass the screening chamber in case of any emergency or maintenance work and divert the sewerage to the existing screening system (the existing screening and wet well must be maintained functionally). In normal operation, the inlet chamber shall be used to divert the sewage to the newly constructed operational screen chambers, which shall be equipped with mechanical bar screens. The Contractor shall design the inlet chamber, screens, drop structures and other associated components at wet weather peak flows. Level sensors shall be provided in the screen chamber upstream and downstream of the mechanical screen for automatic operation. Continuous Flow measurement shall also be done using weir or other methods at the screening chamber. For the collection and automatic transfer of screened material, a belt/screw conveyor shall be provided where the number of screens is more than one. This conveyor belt shall transfer the material to the screening container or disposal truck. The screening material must be transferred to the landfill site on a daily basis. Screening containers of sufficient size to accumulate one-day maximum screening shall be provided with each screen, and the design of the container shall be compatible with the disposal truck currently used by the Client for automatic shifting of screenings.		
REFERENCE DESIGN BOOKS		
• Metcalf & Eddy, Inc. Wastewater Engineering: Treatment and Reuse. • Wastewater Treatment Plants: Planning, Design and Operation, Second Edition by Syed R. Qasim		
• DROP STRUCTURES		
For this project, vortex tangential inlet type drop structures are recommended, however, the Contractor may opt for a cascade baffled drop structure based on hydraulic model testing (mathematical CFD for all structures and physical model for structures wherever value exceeds 40 cusecs, one successful physical model for each type of drop structure). If the Contractor proposes an alternate type of drop structure, they must provide documentation and specifications in their bid to demonstrate that		

the suggested type is proven and meets or exceeds the design, quality, and performance requirements. Moreover, the alternate type has low capital and operational costs, and better performance along with a long-term history and installation case studies internationally in similar projects and installation of the same capacity.

The EPC Contractor is responsible for reviewing the invert levels between the screening system and the main trunk sewer. A drop structure along with an odor control system must be provided between screens and the main tunnel where the drop is greater than 0.5 m, considering the feasibility of construction and odor generation potential, however, drop structures should be strongly considered whenever drop heights exceed 2-3 m. Computational Fluid Dynamics or CFD modeling (the Contractor shall take the approval of software brand and version), and hydraulic physical modeling shall be used in hydraulics design and utilities to optimize the operational efficiency and reliability of the complete system for the selected type, including deep tunnels, integrated diversion works, drop structure, gates, valves, etc.

The drop structure shall be designed to minimize the negative effects of falling flow by:

1. Dissipating energy from the falling flow.
2. Reducing the amount of air entrained by the falling flow and transported into the main tunnel.

DESIGN CRITERIA

The Contractor must validate the proposed design of the drop structure through

the design criteria for the design of a tangential inlet vortex-type drop structure are based on a model-based study (Kennedy, July 1983). The initial drop structure dimensions can be estimated as under:

- Selection of model scale ratio

$$L_r = (Q_p/3.5)^{2/5}$$
- Determination of model values for required de-aeration chamber length

$$LC_m = e^{(0.213 \cdot Q_m + 1.74)}$$
- The determination of dimensions of the drop structure by using model scale ratio L_r is shown in the Table below.

Multipliers used to determine the dimensions of Drop Structure

Multipliers	Description
D1=LR x 0.96	D1= Drop shaft Diameter
D2=Lr x 1.69	D2= Deaeration Chamber Diameter
D3=Lr x 0.84	D3= Deaeration Chamber Outlet Diameter

	LC=Lr x LCm LV=Lr x 4.67 e=Lr x 0.24 Lt=(B-e)/Tan 16.8°	LC= Deaeration Chamber Length LV= Air Vent Location e= vortex opening prior to vertical shaft Lt =Approach Channel Transition Length Where B = approach channel diameter, and e = constricted opening in the vortex generator	The Contractor shall use the specified design criteria mentioned above and peak flows for each disposal station for the optimal operation of the drop structure constructed. For locations where the drop between the screening system and tunnel sewer is low (less than 3m) and has low peak flows, less than 1~2cuses and a selected drop structure is practically not possible due to dimensional constraints, a plunge-type drop structure or drop manholes may be proposed. • VENTILATION SYSTEM A mechanical ventilation system shall be provided for the trunk sewer for the following reasons: • To release gases into the atmosphere in a controlled way to avoid odor problems. • Providing air release to avoid pressurization of the sewer and preventing system blockages that might arise from inadequate or nonexistent ventilation. • To reduce hazards to maintenance personnel by countering the development of a toxic and/or explosive atmosphere. • To promote aerobic conditions and reduce septicity in the sewer by maintaining a supply of oxygen. • To reduce H₂S concentration in the sewer atmosphere. The ventilation system shall consist of induct vents, fan stations, an air treatment system, e-ducts, piping and fittings, and instrumentation. Air entry points and fan stations, along with the air treatment system and educts, will be strategically placed in several locations to effectively remove harmful and unpleasant gases from the system. DESIGN CRITERIA As per the sewage manhole plan of the trunk sewer, the air extraction points using the ventilation fan station along with the odor control system and educts shall be provided at all points mentioned below based on the following minimum design criteria. • At all drop structures
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	• At the bottom of a sudden change in grade from a steeper to a shallower slope • Lateral sewer joints to a Trunk sewer or two or more trunk sewers • Changes in the size of the sewers • Pumping station Other locations shall also be considered based on the software modeling of the ventilation system of the complete trunk sewerage system, along with branch lines. The Contractor must make use of computer modelling for the design of ventilation system through well-known tried and tested computer software (the Contractor shall take the approval of software brand and version) to determine the ventilation requirement and identify all the odor-sensitive areas in the complete trunk sewerage system and an appropriate air treatment system must be installed before releasing air into the sensitive areas, such as buildings, housing areas, or locations, where the ventilation stack is near residential high-rise buildings and could cause odor problems. The Contractor must make sure that for the above-ground installation of Induct vents (air entry point), which should be made of SS316L and contain air intake louvers to facilitate the entry of air into the sewers. The maximum allowable offset distance from the vented sewer or structure to an induct is 20 m. The opening area of the induct shall be sufficient to allow entry of the ventilation air requirement at low velocity to avoid large friction losses. The locations of induct vents should be selected to facilitate maintenance works and be fitted with an air damper to control the volume of air entering the sewer. Fan stations shall be constructed above ground near the manholes (air extraction points). Each fan station shall include pressure centrifugal exhaust fans (a standby option of fan shall be provided separately, depending upon available space). Given the corrosive nature of the gases, all internal components of the fans must be made from stainless steel 316L grade. Additionally, the suction and discharge pipes of the fans should also be made of SS316L. The connection between the fan's suction side and the trunk sewer must be designed by the Contractor to allow for unobstructed air movement, minimizing friction losses and ensuring easy maintenance. For addressing the polluted air produced by drop structures and ventilation systems throughout the gravity sewer network, please see the next section (H. Odor Control System). E-duct vents shall be made of SS316L and at least 10 m tall from the ground level. The height of the e-duct shall be increased at odor-sensitive locations by 3m from the top edge of the building. The offset distance of the e-duct from the fan station shall be limited to avoid large friction losses. The design of the educt shall consider all the anticipated forces like wind load and others. The design criteria for the design of the mechanical ventilation system of the gravity sewer network are provided in the Table below.
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Design Criteria for Calculation of Air Flow of Sewer Fan Station	
Basis	Guidance
Empty Sewer	Size fan to achieve a velocity of 1-2m/s in the empty sewer pipe.
Half-Full Sewer	Determine the extraction rate based on a headspace air velocity that is 50% of the sewage velocity. This will intercept moving air and provide a zone of influence both upstream and downstream of the extraction point.
Velocity Parity	Set the air velocity to be equal to the surface wastewater to reduce turbulence in the sewer air and the rate of escape of odorous gases. To calculate air flow, select the headspace cross-sectional area that either corresponds to ADWF, or the flow rate prevalent when odor complaints if this is happening.
Minimum Air changes	2~5 per hour
The ventilation airflow rate, as outlined earlier, must be at least sufficient to provide 2 air changes per hour.	
• ODOR CONTROL SYSTEM The odor control systems proposed for this project are designed to eliminate foul smells from odorous air so that air can be released into the atmosphere. These gases include, but are not limited to, Hydrogen Sulfide (H₂S), Volatile organic compounds (VOCs), Mercaptans, and Volatile fatty acids. For the treatment of contaminated air generated at drop structures and ventilation systems located at various points of the gravity sewer network, granular activated carbon filters shall be provided; however, the EPC Contractor may also propose a Bio scrubber/gas scrubbing towers or other odor control system as per approval of the Engineer, considering low capital, operation and maintenance cost. If the Contractor proposes an alternate option for the odor control system, they must provide documentation and specifications in their bid to demonstrate that the suggested type is proven and meets or exceeds the design, quality, and performance requirements. Moreover, the alternate type has low capital and operational costs along with a long-term history and installation case studies internationally in similar projects and installation of the same capacity. The scope of supply of the odor control system includes a minimum of an activated carbon dry adsorption system (pressure activated carbon filter), centrifugal high-pressure air fan, electrical controls, air ducting (suction and discharge), activated carbon media, H₂S detector, pressure gauges, and valves. The design of the odor	

control system shall be designed properly so that activated carbon media achieve over 99% removal of hydrogen sulfide (H₂S) molecules. The Contractor is responsible for determining of possible maximum H₂S concentration in the truck sewer under local conditions (worst case scenario). The effluent H₂S concentration shall be below the threshold limit of human detection (0.1 ppm).

The media life must last at least 12 months, and the Contractor is responsible for supplying the necessary media for the first time, as well as for an additional year.

For regular monitoring and maintenance of the carbon filters installed, H₂S gas sensors shall also be provided before and after the activated carbon vessels to monitor the performance of the air treatment system.

DESIGN CRITERIA OF ACTIVATED CARBON FILTERS

The design criteria for the design of the activated carbon odor control system are given in the table.

Design Criteria for Activated Carbon Filters

Criteria	Typical Range
Face Velocity	20 to 100 fpm (0.1 to 0.5 m/s), 50 fpm is selected
Contact Time	3sec for H ₂ S<100 ppm
Pressure Drop	2" WC per foot GAC 1"

DESIGN CRITERIA OF CHEMICAL SCRUBBING TOWERS

The design criteria of the chemical scrubbing towers for the odor control system are given in the table below.

Design Criteria for Chemical Scrubbing Towers

Parameter	Unit	Range/Value
Packing depth	m	1.8–3
Gas residence time in packing	s	1.3–2.0
Scrubbing liquid flowrate	kg H ₂ O/kg airflow	1.5–2.5
Scrubbing liquid flowrate	L/s per m ³ /s airflow	2–3
Makeup water flow	L/s per kg sulfide at pH 11	0.075
Makeup water flow	L/s per kg sulfide at pH 12.5	0.004
pH	Unitless	11–12.5
Temperature	°C	15–40

Caustic usagekg NaOH/kg
sulfide

2–3

DESIGN CRITERIA OF BIO-FILTERS

The design criteria of the bio-filters for the odor control system are given in the table below.

Design Criteria for Bio-Filters

Item	Units	Biofilter	Biotrickling Filter
Oxygen concentration	Parts oxygen/parts oxidizable gas	100	100
Moisture			
- Compost filter	%	50–65	50–65
- Synthetic media	%	55–65	55–65
Temperature, optimum	°C	15–35	15–35
pH	Unitless	6–8	6–8
Porosity	%	35–50	35–50
Gas residence time	s	30–60	30–60
Depth of medium	m	1–1.25	1–1.25
Inlet odorous gas concentration	g/m ³	0.01–0.5	0.01–0.5
Surface loading rate	m ³ /m ² ·h	10–100	10–100
Volume loading rate	m ³ /m ³ ·h	10–100	10–100
Liquid application rate	m ³ /m ² ·d	-	0.75–1.25
Elimination rate (capacity)			
- H ₂ S (in compost filter)	g/m ³ ·h	80–130	80–130
- Other odorous gases	g/m ³ ·h	20–100	20–100
Back pressure, maximum	mm of water	50–100	50–100

SUPER OXYGENATION SYSTEM

Super oxygenation is a high concentration dissolved oxygen technology for preventing anoxic conditions in a variety of water and wastewater applications, using pure oxygen as an alternative to aeration or chemicals.

A side stream of raw wastewater is pumped through a down-flow oxygen contactor, called the Speece Cone, where oxygen gas is introduced. The cone-shaped geometry and the resulting velocity gradient ensure all oxygen is fully dissolved before returning to the pipe or basin.

	Applicability: Flow: 250 gpm - 25,000 gpm D.O.: 30 mg/L - 300 mg/L DESIGN CRITERIA FOR DISPOSAL STATION AND MECHANICAL COMPONENTS This section provides the design criteria and technical requirements necessary for the hydraulic and mechanical design, aimed at ensuring the effective functioning of pump stations and their related hydraulic components. All design work shall be executed in compliance with the specified design criteria, mechanical technical specifications, and drawings. The pump station shall be designed for a sewage (containing solids) peak flow capacity of 688 cusecs with a minimum of ~25% standby capacity of pumps to lift the sewage from the deep trunk sewer discharge to the Pump station outlet channel that will transfer the sewage towards Outfall Channel across Controlled Access Corridor / Bund Road via RCC channels by gravity. A distribution chamber shall be constructed to collect the sewage from the new and old pumping stations at Gulshan-e-Ravi and dispose of it to the outfall structure by gravity. • INLET CHAMBER The sewage inlet chamber shall be designed to collect the sewage from the deep tunnel and direct it to the operational screening channel, or to isolate the pumping station and screen during maintenance or repair activities. DESIGN CRITERIA The inlet chamber design shall prioritize the efficient and safe transport of sewage to the operational screening channels while reducing solids accumulation and ensuring a seamless transition to the screening chamber, thereby preventing excessive turbulence and odor generation. Primarily, the dimensions of the inlet chamber should be determined by the sizes of the connected inlet and outlet channels, ensuring adequate space for the installation of penstock gate valves and facilitating future operation and maintenance. The design must accommodate anticipated peak flow rates, including both average daily flows and peak storm conditions. Additionally, it should aim to minimize head loss within the diversion structure to avert backflow and promote efficient functionality. Design Criteria of Inlet Chamber at Pumping Station <table><tr><th>Description</th><th>Value</th><th>Units</th></tr></table>	Description	Value	Units
Description	Value	Units		

	Material of Construction	RCC	
	Minimum effective detention time	12~15	seconds

- **PENSTOCK GATE**

DESIGN CRITERIA

The design criteria for sluice gate valves shall prioritize reliable and efficient control of sewage flow, strong structural integrity, and effective leak-proof sealing. The thickness of the gate and the configuration of beam frameworks shall be adequate to endure the water pressure and operational forces under double-acting conditions. The mechanism, such as the stem and handwheel, must possess sufficient strength to operate the gate against the expected hydraulic load and friction. The design of the water stop pressure plate is vital for ensuring a secure seal against water flow and must be robust enough to handle pressures while maintaining a tight seal at a minimum water head of 10m. Adhering to standardized design practices and established guidelines (e.g., AWWA/BS/DIN standards) is essential for guaranteeing quality and safety. The valve should be tailored to accommodate the specific hydraulic characteristics of the application, including maximum flow rates, pressure drops, and water depths. Additionally, the design should facilitate straightforward inspection, maintenance, and repair of the valves.

Design Criteria of Penstock Gate Valves

Parameter	Design Criteria
Type of Penstock Gate	Wall-mounted pedestal Type or Channel Type (in case of channel type sluice gate, a recess shall be formed in the RCC channel to avoid reducing the area of water flow)
Size	According to the dimensions of the mechanical screen and the maximum water level
Sealing System	4-side sealing for isolation of flow
Material Construction of	Main plate and frame construction: Stainless Steel SS316 Sealing: EPDM
Sealing design	Double acting
Design head	10 m
Operation	Manual, rising stem
Leakage Rate	Maximum allowable leakage according to BS 7775 for on- and off-seating heads up to and

		including 6 meters – 0.5 liters per minute per meter of seal perimeter																													
	• MECHANICAL BAR SCREEN The EPC Contractor shall comply with the following design criteria and requirements for the design of mechanical bar screens at final disposal station: DESIGN CRITERIA The mechanical bar rake coarse screens shall feature a design that includes a cable-operated cleaning mechanism, utilizing an overhead travelling monorail crane to enable operation across multiple screens. The travelling screen-hoist type will employ a cable-operated screening grab made of SS316, which will be lifted from an overhead rail assembly. One or two cleaning units (determined by the Contractor based on the frequency of cleaning as per the design; the screen should not be more than 50% clogged, and the water level must not increase to a level that reduces upstream velocity less than 0.3m/s at lowest flows, which could result in solid settlement) will be utilized to sequentially clean six screening channels, or as specified in the design. The operation will be fully automated using multiple sensors/limit switches or as per design, with the stainless steel screening grab moving to its designated channel, descending for the cleaning process, ascending afterward, and finally discharging debris at the designated dump truck before proceeding to the next channel. If another type of bar screens (cleaning mechanism: multi- rake chain driven, reciprocating rake (with chain or hydraulic assembly) are proposed, the Contractor must demonstrate in their Bid, through documentation and specifications, that the suggested equipment is proven and meets or exceeds the design, quality, and performance requirements outlined in the design criteria. The Client shall decide the most appropriate type of screening mechanism based on local conditions, cost, quality, reliability, proven design and low operation and maintenance cost. The design criteria for the mechanical bar rack coarse screen are provided in the table below. Design Criteria for Mechanical Bar Screens <table border="1"> <thead> <tr> <th>Parameter</th><th>Unit</th><th>Range</th><th>Selected Value</th></tr> </thead> <tbody> <tr> <td colspan="4">Bar Size</td></tr> <tr> <td>Width</td><td>mm</td><td>10~15</td><td>15</td></tr> <tr> <td>Depth</td><td>mm</td><td>30~38</td><td>30</td></tr> <tr> <td>Clear Spacing between Bars</td><td>mm</td><td>25-50</td><td>40</td></tr> <tr> <td>Slope from vertical</td><td>°</td><td>0~30</td><td>30</td></tr> <tr> <td>Approach Velocity</td><td></td><td></td><td></td></tr> </tbody> </table>			Parameter	Unit	Range	Selected Value	Bar Size				Width	mm	10~15	15	Depth	mm	30~38	30	Clear Spacing between Bars	mm	25-50	40	Slope from vertical	°	0~30	30	Approach Velocity			
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Maximum	m/s	0.6~1.0	0.6~1.0
Minimum	m/s	0.3~0.5	0.3
Maximum Allowable Head loss	mm	150~600	250
Maximum velocity through the screen	m/s	0.9	0.9

The mechanical bar rack screens shall adhere to the design specifications and drawings, capable of handling the peak flow calculated. At least one screen shall be designed as a standby. The stainless-steel SS316 bars used in the manufacturing of the screen shall have a minimum thickness of 15mm and a depth of 30~38mm. The Contractor shall ensure clear bar spacing of 40mm between the bars for efficient waste retention at final disposal station. The screen bar angle from the vertical should be in the range as specified 30° (70° from the horizontal). The depth of the screen channel and the total length of the screen shall be calculated as per the invert level of the trunk sewer. The Contractor must ensure the velocity of water passing through the screen bars does not exceed 0.9m/s to limit the head loss through the screen. The total screen chamber width shall ensure the minimum approach velocity, preferably at all flow conditions (particularly at low flows under present conditions), to avoid the settlement of solids. A proportional weir in SS316 construction shall be designed and installed after each screen to maintain the velocity in the channel during low flow conditions to avoid the settlement of solids at the bottom of the screen channel.

Design Criteria for Proportional Weir

Parameter	Formulas	Units
Flow through a proportional weir	$Q = 1.57 \times C_d \times (2g)^{0.5} \times LH^{3/2}$	m ³ /s
Length	$L = Q / 4.173 (H)^{3/2}$	m
Cd	0.6	

Other configurations, if proposed by the EPC Contractor, must be technically justified with literature, design calculations, and manufacturer recommendations in their Bid. The complete submittal with technical literature and shop drawings shall be submitted by the Contractor for approval from the Engineer.

Level sensors shall be provided in the screen chamber upstream and downstream of the mechanical screen for automatic operation. Continuous Flow measurement shall also be done using weir or other methods at the screening chamber.

The screening material must be transferred to the landfill site on a daily basis. Screening collection trucks of sufficient size to accumulate one-day maximum

	screening shall be provided with the screening system. In case of storage containers, the design shall be compatible with the disposal truck currently used by the Client (WASA-L) for automatic shifting of screenings. • PUMPING STATION The EPC Contractor must adhere to the following design criteria and specifications for a large sewage pumping station, which includes a wet well for sewage collection and pumping, along with a dry well for the installation of sewage pumps. <i>DESIGN CRITERIA</i> The pumping station along with all the components, must be specifically designed to handle sewage water that contains solid particles, rags, fibrous materials, grit, and other potential waste. <i><u>Type of Wet Well</u></i> The wet well, designed for wastewater collection and subsequent pumping via centrifugal non-clogging pumps, must be a proven type suitable for large pumping stations with a maximum capacity of 688 cusec (~20 m³/s) and equipped with 16 high-capacity non-clogging dry pit centrifugal pumps. The EPC Contractor is responsible for selecting the most appropriate wet well configuration for sewage pumping, which may include options such as rectangular configurations with baffle walls and slopes, confined types, or trench designs. For rectangular pumping stations, adequate floor slopes must be incorporated to prevent solid accumulation at the bottom of the wet well. Additionally, the EPC Contractor must ensure that there are no dead ends or corners that could facilitate solid settlement. Provisions for regular cleaning must also be included. However, in any case, the required minimum effective volume must be provided as per design criteria, and the water flow into each pump must remain uniform, steady, and devoid of swirl and entrained air. Furthermore, the design of the pumping station should accommodate variable incoming flow rates throughout the day. The EPC Contractor is required to present a comprehensive comparison, referencing an international pumping station of equal or greater capacity, and to substantiate in their Bid, via documentation and specifications, that the proposed type is validated and satisfies or surpasses the design, quality, and performance standards specified in the design criteria. The following points must be considered: • The required sump volume must be defined to ensure a minimum retention time and to limit the number of pump starts per hour as outlined in the design criteria.
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	• Significant sewage falls should be avoided; thus, the distance between the sewage entrance and the pump inlets must be adequate for air to rise and escape before reaching the pumps. • The energy from falling water should be sufficiently dissipated to prevent excessively high and irregular velocities within the sump, which can be achieved through the use of well-designed baffle walls. • The sewage from the wet well entrance (after the screening system) shall be directed towards the pump inlets in a manner that minimizes swirl. • To prevent the creation of air-entraining surface vortices in the wet well, consider the baffle walls and other design features to eliminate stagnation areas. Additionally, the water depth must be sufficient to suppress surface vortices while avoiding excessive turbulence or large eddies. • It is essential to prevent the accumulation of foul sediment within the sump, steering clear of stagnant areas where sedimentation could occur. Implementing a sloped floor along with fillets or benching can help mitigate sediment buildup. • Scum may build up in the stagnation regions of the wet well and must be removed through pumping. Provision shall be provided for the removal of scum from the wet well. The water level should be periodically lowered to enhance both velocity and turbulence, ensuring that air is not drawn into the pump. <u>Wet Well Volume</u> For the pump stations, all the centrifugal pumps shall be identical and the following design approach shall be applied for the calculation of the volume of the wet well; • The required volume shall be minimized by sequencing pump operations considering fixed speed. • Pumps shall start in sequence as the water level rises due to inflow and stop in sequence as the water level decreases due to outflow. • The last pump to start shall also be the last to stop, ensuring cyclic alternation. The design criteria for the calculation of the effective volume of the wet well are provided in the Table below: Design Criteria for calculation of the effective volume of the wet well <table><tr><th>Parameter</th><th>Unit</th><th>Value</th></tr></table>	Parameter	Unit	Value
Parameter	Unit	Value		

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	Detention time at the average sanitary dry weather flow	mint	5
	Maximum Pump number of starts per hour	Starts/Hr	2*

* The number of starts per hour for an electric motor depends upon the manufacturer data, if a different value is adopted in the design based on pump motor manufacturer recommendations, the EPC Contractor must demonstrate in their Tender/Bid, through documentation and specifications, that the suggested value meets or exceeds the design, quality, and performance requirements outlined in the design criteria.

The effective volume of the wet well shall be calculated by the EPC Contractor using both the criteria as mentioned above and a higher value shall be adopted for the design.

The effective sump volume can then be determined as

$$V(\text{total}) = V_{\text{req}}n + (n-1) \cdot h \cdot S$$

Where S is the plan area of the sump, V_{req} is the total volume required for a single pump operation, and n is the number of operational pumps in the alternating cycle. The start and stop levels of all pumps differ by a constant value h that is determined by the characteristic of the control system. h should be large enough to eliminate accidental pump starts that could be caused by surface waves or imprecise level sensors. A value of 0.2m or higher shall be adopted, however the contract shall provide the reference of adopted value from the practically implemented large size projects of similar capacity. If any other formula is used to derive the wet well volume, the EPC Contractor shall provide the detailed calculation along with the reference.

The design of the wet well and complete pumping station shall also consider the available space at Gulshan-e-Ravi site.

• Hydraulic Design

The sewage pumping station shall be designed to prevent adverse flow conditions and baffle walls and other components shall be provided to avoid vortex formation, pre-swirl, air entrainment, which can disturb the pump performance.

The EPC Contractor shall maintain uniform and steady flow conditions to prevent the pump from operating outside its optimal design parameters, which can reduce hydraulic efficiency. Deviations can also result in heightened noise, vibration, and may cause bearing issues or fatigue failures in pump shafts. Such problems can adversely affect both the performance and lifespan of the pumping system, necessitating careful design and execution to avoid them.

	<i>Excessive pre-swirl</i> Pre-swirl changes the flow conditions at the pump inlet, which results in a change in the relative impeller speed. This, in turn, causes a change in pump performance, which can lead to overloading the motor or reduced pump performance. Excessive pre-swirl can also result in bearing wear and cavitation across the impeller area. Pre-swirl usually originates from an asymmetric velocity distribution in the approach channel, which evolves into a pre-swirl at the pump inlet. <i>Entrained air</i> It is widely known that even minor air entrainment will lead to a clear reduction in pump performance and loss of efficiency; the severity depends upon the quantity of air entrained and the pump type. The expansion of ingested air bubbles within the impeller may result in mechanical imbalance, vibration, and acceleration of mechanical wear. In addition, entrained air leads to increased corrosion. <i>Vortices</i> Unlike excessive pre-swirl, vortices appear locally with higher intensity and are a major hindrance to proper pump operation, resulting in cavitation, uneven load, noise, and vibration. There are several different types of vortices. The commonly known type is the free surface vortex, which can have varying degrees of intensity – from weak surface vortices to fully developed vortices with a continuous air core that extends from the surface into the pump. The dimensions of the pumping station's wet well and other components shall be calculated by the EPC Contractor based on CFD analysis and the “prototype sump hydraulic modelling and testing” for the new pumping station at Gulshan-e-Ravi. Ultrasonic level sensors shall be provided in the wet well to measure the water level continuously and for the automatic cyclic operation of the pumps. One additional sensor shall also be provided to enhance the accuracy of reading and to provide backup in case of malfunctioning of the sensor. In the dry well, other pump accessories should be provided, such as gate valves, swing check valves, pressure gauges, pressure transmitters, flow meters, piping, EOT cranes, transient control equipment if required and all other items required for the correct operation of the pumping station. • Model Testing & CFD Analysis As per the recommendations provided in “Pump Intake Design by Hydraulic Institute”, physical model testing shall be conducted by the EPC Contractor for the new pumping station at Gulshan-e-Ravi. In addition, a CFD analysis shall also be performed to simulate the flow behavior at various flow conditions.
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	<i>Model Study</i> A properly conducted physical model study is required to identify unacceptable flow patterns at pump suction for given sump or piping designs and to derive acceptable intake sump or piping designs. <i>Model Objective</i> Adverse hydraulic conditions that can affect pump performance include; free and sub-surface vortices, swirl approaching the pump impeller, flow separation at the pump bell, and non-uniform axial velocity distribution at the suction. A properly conducted physical model study shall be used to derive remedial measures, if necessary, to alleviate these undesirable flow conditions. The objective of a model study is to ensure that the final sump or piping design generates favorable flow conditions at the inlet to the pump. <i>Model Similitude and Scale Selection</i> Models involving a free surface are operated using Froude similarity since the flow process is controlled by gravity and inertial forces. The Froude number, representing the ratio of inertial to gravitational forces can be defined for pump intakes as; $F=u/(gL)^{0.5}$ Where: u=Average axial velocity (such as in the suction bell) g= gravitational acceleration L=a characteristic length (usually bell diameter or submergence) The choice of the parameter used for velocity and length is not critical, but the same parameter must be used in the model and prototype when determining the Froude number. For similarity of flow patterns, the Froude number shall be equal in model and prototype: $Fr=F_m/F_p=1$ Where m, p, and r denote the model, prototype, and the ratio between model and prototype parameters respectively. In modeling a pump intake to study the potential formation of vortices, it is important to select a reasonably large geometric scale to minimize viscous and surface tension
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	scale effects. Also, the model shall be large enough to allow visual observations of flow patterns, accurate measurement of swirl and velocity distribution, and sufficient dimension control. Preferably a model scale ratio of 1:10 shall be adopted for model construction. Model Scope Selection of the model boundary is extremely important for the proper simulation of flow patterns at the pump. As the approach flow non-uniformities contribute significantly to the circulation causing pre-swirl and vortices, a sufficient area of the approach geometry or length of piping has to be modeled, including any channel or piping transitions, bends, bottom slope changes, control gates, expansions, and any significant cross-flow past the intake. All pertinent sump structures or piping features affecting the flow, such as screens and blockage due to their structural features, trash racks, dividing walls, columns, curtain walls, flow distributors, weirs and piping transitions must be modeled. Special care should be taken in modeling screens; the screen head loss coefficient in the model shall be the same as in the prototype. The head loss coefficient is a function of the screen. In general, comply with all the requirements of the referenced standard. Acceptance Criteria The acceptance criteria for the model test of the final design shall be as per the criteria given in "Pump Intake Design by Hydraulic Institute" or any other internationally acceptable standard as per Employer approval. Type of Pumps • The centrifugal pumps shall be of a non-clogging type for dry pit installation, featuring a cardan shaft with vertical electric motors at floor level. All the sewage pumps shall be single-stage, volute casing rugged models with durable construction, featuring end suction, radial discharge and having an inspection plate in volute. • The pumping station dry well will be equipped with a total of 16 centrifugal sewage pumps, 13 of which are operational and 3 on standby. The flow rate of each pump is 53 cusecs. The number of operational pumps with different flow conditions under current conditions are shown below:
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Flows at Present Conditions 2025			No. of Operational Pumps, each with a capacity of 53 cusec
Minimum Dry Flow	56.2	cusec	1
Average Flow	137.6	cusec	2~3
Peak Flow	266.8	cusec	5
Peak Flow with Storm water	391.2	cusec	7

- It is significant to highlight that the EPC Contractor is permitted to determine the number of pumps based on their calculations, provided that the overall life cycle cost of the pumps is reduced.
- If a higher capacity pump is selected by the Contractor, the selection of pumps should consider all sewage flow scenarios of the present conditions, including minimum, average, and maximum peak flows.
- The chosen quantity of pumps must also take into account the number of starts per hour for the higher KW electric motor, which should not surpass the limits set by the motor manufacturer recommendations.
- The pumps shall be proposed with a minimum pump efficiency of 80% or higher at peak sewage flow conditions at the duty point.
- The EPC Contractor must determine the total pump head needed at the station, ensuring it does not exceed 26m, including all safety margins.
- A minimum of 15% safety margin shall be built in the total head of the pumps.
- The Pump Characteristic curve must comply with ISO 9906:2012 1U in respect of tolerance in Flow, Head and Pumps Efficiency. The velocity inside the pressurized pipes shall not exceed 2m/s.
- The velocity in the suction pipe shall not be greater than 1.7 m/s.
- The NPSH_{available} shall be at least 30% greater than NPSH_{required}.
- All the fittings bends shall be medium or long radius; short radius bends will not be acceptable.
- The overall design of the system shall be optimized for reduced energy consumption and fewer losses, and promote gravity flow.
- The lowest and minimum water levels in the wet well will establish the static head, determined by the invert level of the trunk sewer at the disposal station.
- The EPC Contractor must ensure the correct invert level of trunk sewer while designing the pumping station and pump head.
- The pumps shall lift the sewage to the outfall disposal channel for the ultimate disposal of sewage to River Ravi.

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	• The EPC Contractor is responsible for conducting hydraulic analysis and assessing friction losses in the pressure piping system to determine the total dynamic head of the pump using established formulas such as the Hazen-Williams or Darcy-Weisbach equations. • Additionally, a detailed transient analysis of the pump and pressure pipes will be performed, which involves simulating and analyzing the dynamic variations in pressure and flow within the piping system over time, especially during sudden flow changes or disturbances, or sudden pump trip due to power failure. This analysis is essential to prevent problems such as pressure surges, commonly referred to as water hammer, and valve slams, which can lead to damage to pipes and infrastructure. Suitable surge suppression mechanisms and devices will be put in place. • The EPC Contractor shall utilize computer software for hydraulic and transient analysis, including AFT Mercury and AFT Impulse or equivalent. <i>DESIGN CRITERIA</i> <i><u>Pipe Nominal/Outer/Inner Diameters</u></i> For the hydraulic calculations of pumps, ductile iron pipe with cement mortar lining as per ANSI/AWWA C150/A21.50 and ANSI/AWWA C151/A21.51 shall be used. Cement mortar lining as per AWWA C205 shall be used. The pipe's inner diameter is the outer diameter of the pipe minus twice the pipe thickness and cement mortar lining thickness. <i><u>Velocity (V)</u></i> The velocity of water in the pipeline is calculated based on the continuity equation given as under $V = Q / A$ Where: V = Flow Velocity in Pipeline (m/sec) Q = Flow rate per pipeline (m³/sec) A = Pipeline effective Area ($\pi D_i^2/4$) Where D_i is Pipeline Internal Diameter (m)
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	Velocity Head (Hv) The velocity head is calculated based on the following Formula $H_v = V^2 / 2g$ Where Hv = Velocity Head (m) V = Velocity in Pipe (m/sec) Friction Losses (Hf) Friction losses in the pipeline are calculated based on the Hazen – William formula given as under $H_f = 10.666 \times C^{-1.85} \times D^{4.87} \times Q^{1.85} \times L$ Where: Hf = Friction Losses (m) C = Hazen William Coefficient (120 -Taken for Cement Mortar Lining) D = Pipe Diameter (m) Q = Flow Rate (m³/sec) L = Pipe Length (m) Minor Fitting Losses (Hm) Fitting losses are calculated using the following formulas; $H_m = K \times V^2 / 2 g$ Where: Hm = Minor Fitting Losses (m) K = Fitting Loss Coefficient V = Velocity in Pipe (m/s) g = Acceleration due to gravity (m/sec²)
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Fitting loss coefficient		
Fitting Head Loss Coefficient (K)		
Description	Values	Unit
90-degree Bends	0.3~0.5	Unitless
45-degree Bends	0.2	Unitless
Tee (Run to Branch)	1.5	Unitless
Tee (Run to Run)	0.5	Unitless
Butterfly Valve	0.5	Unitless
Swing Check Valve	2.3	Unitless
Reducer	0.19	Unitless
Entrance	0.5	Unitless
Exit	1.0	Unitless
Total Dynamic Head (H_t)		
The total dynamic head of a system is the sum of static head, velocity head, and friction losses in a pipeline, given as under		
$H_t = H_s + H_v + H_f + H_m$		
Where:		
H _t = Total Dynamic Head (m)		
H _s = Static Head (m)		
H _v = Velocity Head (m)		
H _f = Friction Head (m)		
H _m = Minor Fitting Losses (m)		
• Ventilation and Odor Control System		
All the odorous and harmful gases received at the disposal station shall be collected and treated through an odor control system based on a minimum of activated carbon vessels and then released to the atmosphere. However, the EPC Contractor may also propose a Bio scrubber/gas scrubbing towers, superoxygenation cone technology or other odor control systems as per approval of the Engineer, considering low capital, operation, and maintenance costs. If the Contractor proposes an alternate option for the odor control system, they must provide documentation and specifications in their bid to demonstrate that the suggested type is proven and meets or exceeds the design, quality, and performance requirements. Moreover, the alternate type has low		

capital and operational costs along with a long-term history and installation case studies internationally in similar projects and installation of the same capacity.

REFERENCE DESIGN STANDARDS

- American National Standard for Pump Intake Design by Hydraulic Institute ANSI/HI-9.8
- Design Criteria - WASA Lahore
- Design Recommendations for pump stations with large centrifugal wastewater pumps by Flygt ITT Industries

DESIGN CRITERIA OF DISTRIBUTION CHAMBER

A distribution chamber is provided downstream of both the new and old pumping stations, designed to collect sewage from these stations by gravity and channel it to the Outfall Channel across Controlled Access Corridor / Bund Road. Additionally, it includes provisions for isolation via penstock gate valves and a bypass to the cantonment drain (spill channel).

The chamber is gated, with control gates at:

- Conduit openings
- Spill channel intake
- Outlets from both pumping stations

DESIGN CRITERIA

The distribution chamber design shall prioritize the efficient and safe transport of sewage to the RCC concrete channels towards the outfall at Bund Road while reducing solids accumulation and ensuring a seamless transition, thereby preventing excessive turbulence and odor generation. Primarily, the dimensions of the distribution chamber should be determined by the sizes of the connected inlet and outlet channels, ensuring adequate space for the installation of penstock gate valves and facilitating future operation and maintenance. The design must accommodate anticipated peak flow rates from both the existing and new pumping stations, including both average daily flows and peak storm conditions. Additionally, it should aim to minimize headloss within the diversion structure to avert backflow and promote efficient functionality.

Design Criteria of Distribution Chamber

Description	Value	Units
Material of Construction	RCC	
Minimum effective detention time	12~15	Seconds

	• Maximum velocity in the chamber: 1.0 m/s (to ensure uniform flow distribution) • DESIGN CRITERIA OF OUT FALL CHANNEL The design of the distribution chamber and downstream RCC discharge channels to transport the combined sewage to the outfall head regulator at the Bund Road shall consider the available hydraulic gradient for gravity flow. DESIGN CRITERIA The EPC Contractor must take into account the following considerations when designing the distribution chamber and RCC discharge channel levels: • Assess the current invert levels of the RCC channels from the existing pumping station and ensure that the same gradient is maintained to support the sewage disposal from the existing pumping station. The design of the new structures should not interfere with the hydraulic and power requirements of the existing system. • At Junction of Main Boulevard Gulshan e Ravi and Bund Road, three RCC buried culverts have already been constructed to allow sewage to flow beneath the Roadway; the Contractor is required to verify the invert levels of these buried RCC culverts and align their design with the existing hydraulic gradient to ensure gravity flow. The inside dimensions of each culvert are 3.5 x 3.094 m. • A transition chamber will be built at the junction of the existing culverts and the new RCC channels from the distribution chamber, provided that the channels are designed with varying dimensions. • The sewage velocity inside the channels shall be in the range of 1~2 m/s. The velocity at the lowest flows at the time of project commissioning shall not be less than 0.6 m/s. • Maximum recommended velocity: 2.74 m/s (9 ft/s), • Preferred velocity at peak flow: 1.75 m/s – 2.0 m/s • For velocities exceeding 2.74 m/s, special approvals from the competent authority will be required. • Sharp bends and corners shall be avoided to reduce the friction losses. All entrances, exist, orifices, bends, etc., losses shall be calculated to estimate the sizes and slope of the channels. FRICION LOSSES IN OPEN CHANNEL The Manning formula is an empirical formula estimating the average velocity of a liquid flowing in a conduit that does not completely enclose the liquid, i.e., open channel flow. All flow in so-called open channels is driven by gravity.
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$$V = 1/n (R^{2/3}) (S^{1/2})$$

Where:

V is the cross-sectional average velocity (m/s)

n is the Manning coefficient. Units of n are often omitted, however, n is not dimensionless, having units of: (s/[m^{1/3}]). For concrete channels, a value of 0.013 is adopted for design.

R is the hydraulic radius (m)

S is the slope of the hydraulic grade line or the linear hydraulic head loss, which is the same as the channel bed slope when the water depth is constant. ($S = hf/L$).

Coefficient of Roughness

The Manning's roughness coefficient (n) for concrete varies depending on surface finish quality.

- Standard roughness coefficient: 0.013
- Recommended roughness coefficient: 0.015 (to account for long-term surface deterioration due to erosive flows).

Values for Manning's coefficient for concrete ranges

#	Finish Quality	Minimum	Normal	Maximum
1.	trowel finish	0.011	0.013	0.015
2.	float finish	0.013	0.015	0.016
3.	finished, with gravel on bottom	0.015	0.017	0.020
4.	unfinished	0.014	0.017	0.020

DESIGN CAPACITY

The design discharge for each component is:

- Outfall Concrete Channel: Approx. 19.82 m³/s (700 ft³/s) – after new pumping station to distribution chamber
- Outfall Concrete Conduit: Approx. 29.74 m³/s (1050 ft³/s) –after distribution to chamber to outfall head regulator at Bund Road (combined flow of new and old pumping stations)

To accommodate these flows, the concrete conduit will be designed as a three-conduit system, with each conduit handling 9.91 m³/s (1/3rd of total discharge).

• DESIGN CRITERIA OF OUTFALL CHANNEL

The trunk sewer system will discharge approx. 19.82 m³/s (700 ft³/s) of sewage into the Gulshan-e-Ravi Disposal Station, where it will be pumped into the outfall disposal drain, leading to River Ravi.

Since the existing disposal station already discharges approx. 10 m³/s (350 ft³/s), the combined discharge capacity of the new and old stations will be 29.82 m³/s (1050 ft³/s).

Existing Outfall Drain and Required Upgrades

The current earthen channel is designed for 9.91 m³/s (350 ft³/s). To accommodate the increased flow, the channel will be remodeled for a capacity of 29.74 m³/s (1050 ft³/s).

- Current corridor width: 22-24 m
- Existing drain width: 16 m
- Total channel length: 1,550 m
- Upstream bed level: 205.0 m.a.s.l
- Maximum flow level: 206.35 m.a.s.l
- Bank top level: 206.65 m.a.s.l

Since the channel passes through densely populated areas, it cannot be widened. Instead, a rectangular brick-walled channel will be constructed within the existing corridor. To improve environmental conditions, 3.0 m wide roads will be built along both banks.

According to open channel hydraulic Volume by Vente Chow, the Manning's 'n' value for the outfall disposal channel varies between 0.011 and 0.018.

For design purposes, n = 0.016 has been selected to account for long-term flow resistance and sediment accumulation.

#	Bricks	Manning' n		
		Minimum	Normal	Maximum
1	Glazed	0.011	0.013	0.015
2	in cement mortar	0.012	0.015	0.018

GRADIENT OF THE DRAIN

The bed slope is based on a hydrographic survey conducted by PMC through a third-party surveyor.

- Cross-sections observed every 25 meters along 1 km length

	- Calculated bed slope: 0.0004 (1 in 2500). Free board - Minimum freeboard required: 0.3 m - Brick walls will be installed along both banks for safety. - Greenery (trees/bushes) will be planted between the road and channel walls to help absorb odor and pollutants. DESIGN OF CHANNEL Using Manning's formula, the following parameters have been calculated: - Velocity: 1.38 m/s (4.52 ft/s) - Depth of flow: 1.35 m - Bed width: 16 m - A fall of 2.2 m is located 375 m from River Ravi, ensuring smooth flow transition. DESIGN OF HEAD-REGULATOR OUTFALL DISPOSAL CHANNEL The intake of the outfall disposal channel features a head regulator equipped with two vertical gates, which are in poor condition and require replacement. - New discharge capacity: 29.74 m³/s (1050 ft³/s) - Upgraded head regulator: Three manually operated vertical gates - Structural components: - Two piers and two abutments - Service bridge with steel/concrete railing for O&M access - Stop-log grooves at upstream and downstream ends for gate maintenance This remodeled regulator will effectively manage flow distribution, ensuring efficient sewage disposal into River Ravi while providing maintenance access for future upgrades. DESIGN OF HEAD-REGULATOR OUTFALL DISPOSAL CHANNEL The following formula will be used for design of head regulator components such as: flow width, glacis, stilling basin, stone apron, cut of walls, etc. The weir discharge capacity is calculated applying the following standard formula: $Q = B'C \sqrt{(2gH_e^3)}$
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	Where, C = Discharge coefficient B' = Effective width He = Head over the crest The effect of piers and abutments on the hydraulically effective crest width and thus on the weir discharge capacity is estimated using the following relationship: $B' = B - 2(n \times k_p + k_a) \times H_e$ Where, B/ = Effective width B = Clear waterway width n = Number of piers k p = Pier contraction coefficient k a = Abutment contraction coefficient He = Head above crest level The piers and abutments cause side contraction of the overflow, thus reducing the effective width of the weir/spillway, also depending on the head over the crest. The contraction coefficients Kp and Ka are affected by the shape of the pier nose and abutment shape respectively. For Broad crested weir Kp=0.01 and Ka=0.10 will be applied. DESIGN OF STILLING BASIN A stilling basin shall be arranged and designed to dissipate largely the hydraulic energy generated by the drop from the weir crest and thus avoid excessive channel and banks erosion downstream of the weir structure. To ensure that the hydraulic jump is maintained within the stilling basin for the entire range of channel discharges the stilling basin with end sill is selected. The length of the stilling basin shall be designed according to common design approaches such as that reported by “USB”. Accordingly, the minimum length of the stilling basin is 5 times the difference of the conjugated depth:
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	$LSB = 5 \times (h_{2req} - h_1)$ Where, LSB = Length of stilling basin h_1 = Conjugated depth at begin of hydraulic jump h_2 = Conjugated depth at end of hydraulic jump <i>WEIR SUB-SURFACE FLOW</i> Excessive hydrostatic head across a regulating structure can induce seepage through the underlying subsoil, leading to uplift pressures beneath the structure. These uplift pressures, if not properly accounted for, can compromise structural stability, increasing the risk of soil failure, piping, or structural displacement. To determine the uplift pressure distribution under steady seepage conditions, Khosla's Theory will be applied. The percentage pressure at critical key points will be computed using Khosla's curves, ensuring accurate pressure distribution assessment. For complex structural profiles, corrections for floor thickness and mutual interference of piles will be required to achieve a realistic uplift pressure profile. <i>Seepage Control Measures</i> To prevent seepage-related structural failure, the following measures will be implemented: <i><u>Cut-off Barriers:</u></i> • Steel sheet piles, stone walls, or slurry trenches will be installed at both upstream and downstream ends of the structure. • These barriers mitigate soil slipping and structural failure caused by scour holes and seepage-induced erosion. <i><u>Exit Gradient Control:</u></i> • Seepage forces generated by water flow through subsoil exert pressure on soil particles. • If the seepage force exceeds the critical exit gradient, it can lead to soil displacement, piping, and eventual structural failure. • Khosla's Theory of Exit Gradient will be used to determine appropriate countermeasures to prevent piping beneath the structure.
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	By incorporating these seepage control measures, the structural integrity and longevity of the regulating structure will be ensured, preventing failure due to excessive uplift pressure and seepage-induced soil movement. DESIGN OF STORM DRAIN INLETS <i>Types of Storm Drain Inlet along Streets/Road</i> The primary function of storm drain inlets is to intercept excess surface runoff and channel it into the drainage system, thereby preventing surface flooding. These inlets are typically installed along streets, where they efficiently collect and convey runoff while minimizing disruption to vehicular and pedestrian traffic. Since streets are primarily designed for vehicular movement, storm drain inlets must be strategically placed and designed to ensure they do not interfere with traffic flow, pedestrian movement, or urban infrastructure. <i>Design Criteria for Inlets on Streets/Roads</i> The following guidelines shall be followed for the design and placement of inlets in urban streets and roads: • Minimum Transition for Depressed Inlets: A minimum transition length of 3.05 meters (10 feet) shall be provided for depressed inlets to facilitate smooth flow transition. • Recessed Inlets Placement: Recessed inlets shall be designed so that they do not obstruct pedestrian movement or interfere with the intended use of sidewalks. • Capacity of Recessed Inlets: The capacity of a recessed inlet placed on a graded surface shall be calculated similarly to that of a non-recessed inlet of the same type. • Pedestrian and Bicycle Safety Considerations: The design and placement of inlets shall be evaluated in relation to pedestrian and bicycle traffic to ensure safety and accessibility. • Compliance with Design Standards: Inlet design and location must adhere to the criteria established in Section III of this manual.
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	• Standard Inlet Lengths: Inlet lengths shall be provided in 1.22-meter (4-foot) increments for design consistency. • Curb Inlet Openings and Safety Measures: Curb inlets with vertical openings larger than 0.20 meters (8 inches) must be equipped with grates to prevent hazards. • Grate Inlets Adjacent to Curbs: Grate inlets located adjacent to a curb must be combination inlets, ensuring efficient water intake and debris control. <i>Types of Storm Drain Inlets</i> Storm drain inlets are classified into three major types, each with different design variations: <i>Type A: Inlet in Sumps</i> Varieties: • Curb Opening • Grate • Combination (Grate & Curb Opening) • Drop Inlet • Drop Inlet with Grate Covering Type B: Inlet on Grade without Gutter Depression Varieties: • Curb Opening • Grate • Combination (Grate & Curb Opening) <i>Type C: Inlet on Grade with Gutter Depression</i> Varieties: • Curb Opening • Grate • Combination (Grate & Curb Opening)
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Each inlet type has specific advantages and disadvantages, depending on site conditions, hydrology, and urban requirements.

Under this project, combination inlets will be used as the standard inlet type due to their high efficiency in capturing runoff. However, at Lakshami Chowk, only grate-type inlets will be installed to accommodate local conditions.

Allowable Depth of Gutter Flow

The maximum permissible depth of gutter flow is determined based on the curb height and is outlined in the table below.

Under no circumstances should the depth of gutter flow exceed the specified limits during the design runoff event to prevent street flooding and ensure safety.

By following these design principles, the project will implement an effective stormwater drainage system, ensuring flood prevention, urban resilience, and safe infrastructure for pedestrians and vehicles alike. The allowable depth of gutter flow is a function of the height of curb and is given in the table below.

Allowable Depth of Gutter Flow

Height of Curb		Maximum Depth of Flow	
(in.)	(mm)	(ft.)	(mm)
3	75	0.200	60
4	100	0.267	80
6	150	0.400	120
8	200	0.450	135

TECHNICAL SPECIFICATIONS OF MECHANICAL ITEMS

GENERAL REQUIREMENTS

The following are the general and minimum requirements for mechanical installations at the Sewerage System from Larech Colony to Gulshan-e-Ravi, Lahore. This document shall be examined in relation to the design criteria for mechanical components.

The Contractor shall be responsible for providing detailed designs of the installations, ensuring full coherence of all the equipment installed, and complying with the present minimum quality requirements.

If the equipment not mentioned in this section is offered, the Contractor shall with his Bid demonstrate through submission of documentation and specification that the proposed equipment is of equal or better quality as per the project requirements.

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	New offered equipment should have higher energy efficiency and quality than the original offer. All the equipment, pumps, pipes, valves, fittings, screens, penstock gate valves, instruments etc shall be products of well-established firms that are fully experienced and qualified in the manufacturing of the equipment to be furnished. The equipment shall be designed, constructed and installed in accordance with the best practices and methods and shall comply with these SPECIFICATIONS as applicable. The equipment design, workmanship and general finish shall be of sound quality in accordance with good engineering practice. Design shall be robust and rated for continuous service at the specified duties under the prevailing operational site condition. <u>SUITABILITY OF DESIGN AND EQUIPMENT</u> During the selection and / or design of equipment and installations, particular attention should be paid to the following: • Safety of operation and easy maintenance • Well-proven and reliable components • Ability to withstand the service conditions • Inaccessibility of Vermin, dust, and humidity • Precautions to minimize corrosion • Spare parts availability • Service availability in Pakistan • Minimization of Noise • <i>Control of ill-smelling, and odor and maintaining health conditions</i> • <i>Alterations of design and make of offered equipment after the contract has been signed, can only take place after the consent of the Engineer.</i> <u>Environment</u> <i>All equipment shall be rated for ambient temperatures, relative humidity, and altitude at the Site.</i> <i>Except where otherwise specified, all equipment supplied shall withstand, without developing any defect, the following climatic conditions of Lahore:</i> • <i>Ambient Temperature: Maximum: +45°, Minimum: +5°C</i> • <i>Outdoor Temperature: Maximum: +50°C, Minimum: 0°C</i> • <i>Relative Humidity: Up to 97%</i> • <i>Altitude: 215-220 meters above sea level</i>
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	<i>The Contractor shall in his design and selection of equipment take into consideration under which conditions each item shall be in operation.</i> Equipment placed in the open (out-door) shall be designed for solar radiation-induced temperatures and stormwater or in some way protected from direct Sunlight/stormwater. Equipment placed in or near wastewater /sewage shall be efficiently protected against intrusion of humidity and water. Equipment placed where the risk of condensation prevails shall be provided with drainage holes placed in the lower part of the equipment. Equipment shall moreover be designed to prevent the intrusion of insects and smaller reptiles. <u><i>Country of Origin</i></u> The country of origin of all mechanical equipment like MTBM, pumps, screens, valves, fans, odor control etc. shall be USA or Western Europe or origin manufacturing equipment's meeting or exceeding the quality and standards maintained by USA or Western Europe. The manufacturers from the above-mentioned origins are also acceptable which have same standards and having manufacturing/assembly facility in China/Pakistan subject to the prior approval of the Engineer/ Employer. It will be mandatory for the Contractor to get the approval of all the submitted data and specifications for equipment from the Employer prior to any procurement. Other than MTBMs, the manufacturer must have an authorized dealership distributor in Pakistan and must have experience of providing after-sale services. <u><i>Third Party Inspection of Imported Items</i></u> Before dispatch of major imported equipment from manufacturer premises, Contractor shall submit a third party inspection report to conform the specifications as described in bill of quantity and technical specifications. Third party inspection report should preferably be submitted from international company having same type of experience as approved by the Engineer. <u><i>Manufacturer Warranty</i></u> The equipment and its accessories shall be warranted for a period of 5 years from the date of commissioning of the equipment. The Contractor shall organize manufacturing warranty certificates from the manufacturer in favour of the Client that the equipment will operate satisfactorily and the performance of the equipment when operating under normal conditions shall not be less than the guaranteed values. The manufacturer of equipment shall further
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	guarantee that during the warranty period, they will replace all defective materials, accessories, etc. furnished or installed under the contract, free of cost to the purchaser/Client. <u>Name Plates and Labels</u> Each main and auxiliary item shall have a nameplate, permanently attached to the equipment at a conspicuous position. Upon these shall be engraved, in English language, the manufacturer's name, type and serial number of equipment, details of the loading and duty at which the item has been designed to operate, and such diagrams as deemed necessary. <u>Special Tools and Test Equipment</u> Contractor shall supply a complete set of special tools and test equipment necessary for the maintenance or testing of any part of the equipment, to be carried out during the life of the equipment. The tools and test equipment shall not be used for the erection of the equipment and shall be handed over in a new and unused condition, except that the Employer may call upon the Contractor to prove their effectiveness. The tools and test equipment shall be boxed separately in suitable containers, duly marked or labeled with their contents. <u>Storage of Material</u> Contractor shall provide weatherproof lock-up sheds for the safe storage and custody of material for the works, and shall remove such sheds and other materials and repair the disturbed surfaces upon completion to the satisfaction of the Engineer. <u>Safety</u> The Contractor shall ensure that all designs and equipment for which he is responsible are safe. Nothing in these specifications shall remove the Contractor's obligation from drawing the attention of the Employer to any feature of the works, which is not consistent with safety. The installation layout and equipment design shall not allow any item of equipment to be so positioned, that danger to operating personnel could arise during normal operation and maintenance. Particular attention shall be paid to the position of air vents and rotating machinery. All rotating shafts, couplings, gears, flywheels, belt/chain drives or other moving parts shall be fully guarded. Guards shall be designed to provide ready access to bearings, grease points, and other checkpoints and to allow safe routine observation and
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	servicing to be executed without the need to dismantle any part of the structure or equipment. <u><i>Drawings and Documents</i></u> • The Contractor shall prepare and submit for approval, five (5) sets of his quality assurance plans giving details of the time schedule covering various phases e.g. Preparation of shop drawing, procurement of material, manufacturing time, assembly, inspection testing, delivery to the site etc. • The Contractor shall submit five (5) copies of the drawings, submit for approval before commencing with shop manufacture. These drawings shall include general arrangement drawings of the pumps, coupled detailing of the motor including dimensions, nozzle details etc., Cross-sectional assembly drawing of pumps giving fabrication details, component details and bill of materials with the material of construction etc., foundation drawings giving loading data and grouting details, etc. motor dimensional drawings. • The Contractor shall submit the technical data sheet and other literature like selection criteria, performance curves, capacity, pressure, efficiency, power consumption, weight, static and dynamic loads and other relevant data. • After approval of drawings and other documents, the Contractor shall submit five (5) copies of all relevant material test reports, manufacturer's test certificates, hydrostatic test certificates, performance test certificates, etc. as asked in the specification for inspections and testing. • The Contractor shall submit five (5) numbers of bound copies of instructional manuals for each mechanical equipment for installation, operation and maintenance. The manuals should include drawings and performance curves, assembly and disassembly procedures, parts replacement, etc. the manual shall be got approved by the Engineer. <u><i>Coordination</i></u> It is the responsibility of the Contractor to ensure full coherence between the equipment delivered according to the present mechanical specifications and the specifications for electrical and civil works. The Contractor shall also be responsible for all sub-Contractors and suppliers of equipment and materials. No direct formal communication between the Employer and the sub-Contractors will be accepted.
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	It is the responsibility of the Contractor to ensure those sub-Contractors and suppliers get all the relevant information of the present specifications. The Contractor shall appoint and provide an experienced mechanical and electrical engineer to monitor and coordinate all aspects of the mechanical and electrical works. <u><i>On site Manufacturing</i></u> It is the obligation of the Contractor to provide all installations within the working area as he considers necessary for on-site fabrication and shaping of materials. The working site may be used as intermediate storage for material at the Contractor's own risk and at his own cost. It shall be noted, that the Contractor's obligations to take the necessary safety precautions shall apply to the working camp area as well. <u><i>Erection</i></u> The Contractor shall make his own arrangements for the unloading of equipment and materials supplied and shall be responsible for any damage that occurs. The Contractor shall at his own expense provide all tools, meters, gauges, temporary provisions as well as skilled and unskilled labour for the erection of the mechanical installations so that it can be installed completely and in good working order. For the proper installation of all mechanical equipment under this Contract an authorised representative of the manufacturer and its service organisation shall approve the installation and sign together with the Contractor a statement with such approval. A form of such statements will be provided by the Engineer. <u><i>Foundations, Building Work and Setting of Machinery</i></u> • The Contractor shall ensure that the positions of foundations for machinery/equipment, RCC foundations, holding-down bolts and the setting of machinery are positioned in accordance with the approved machinery drawings. • The Contractor shall, upon receipt of the necessary approved drawings for the machinery, carry out excavation work and the construction of all the necessary foundations and bases for the various items of the system, including the forming of holes and chases for pipework, steelwork, cabling, conduit, rag bolts and where necessary, the building in of foundation bolts and sundry items of system, all following the drawings. Spaces shall be left between the concrete and bedplates etc. for grouting. • The Contractor shall provide all necessary templates for fixing the positions of bolt holes, etc. • The machinery shall be mounted on flat steel packing of a thickness selected to take up variations in the level of the concrete
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	foundations. The packings shall be bedded by chipping or grinding of the concrete surface. • Only one packing of selected thickness shall be used at each location, which shall be adjacent to each holding-down bolt. The number of shims shall not exceed two at each location and the thickness of each shim shall not exceed 3 mm. • The machinery shall be aligned, levelled and pulled down by the nuts of the holding down bolts with a spanner of normal length, and no grout shall be applied until the machinery has been run and checked by the Employer for stability and vibration. • The Contractor shall clean the concrete and grout up after the pumps, motors, girders, etc. have been finally fixed and packed up. <u>Location Alignment</u> Where separate items of the system, such as motors, couplings, gearboxes and similar items depend upon correct alignment for satisfactory operation, then each and every item shall be positively located in its correct operational position by means of dowels, locating pins, fitted bolts or other approved means to ensure that correct re-alignment can be easily achieved when reassembling the items after removal for overhaul. <u>Lubrication</u> • Equipment shall remain adequately lubricated for its continuous operation as per OEM recommendation. Lubrication systems shall not require attention during start-up or shutdown and shall not waste lubricants. • Lubrication facilities shall be convenient and accessible. Oil drains and fill openings shall be easy accessible from the normal operating area or platform. Drains shall be located so as to allow for collection of waste oil into containers without removing the equipment from its normal position. • Lubricants of the type recommended by the equipment manufacturer shall be provided in sufficient quantities to fill all lubricant reservoirs and to replace all consumption during testing, start-up and operation prior to acceptance of the equipment by the Engineer. • Furthermore, a list of recommended lubricants and their equivalents shall be included in the operation and maintenance instructions. The lubricants recommended shall be available in Pakistan. Sufficient lubricants shall be supplied for operation of the plant during the entire defect's notification period. The quantities shall be based on operating hours of 24 hours per day.
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	SCOPE OF SERVICES <i>General</i> The workmanship and the materials covered by this section shall include the supply, installation, testing, and commissioning of all mechanical equipment which includes pumps, piping, fittings, supports, valves, mechanical bar rake screen, and overhead crane, etc. The Contractor shall supply, transport, deliver, install, connect, commission, and hand over all Mechanical Equipment, and materials specified in the specification, drawing, and bill of quantities in a clean complete, and in detailed working condition. All materials and equipment shall be obtained from a reputable manufacturer, local agents shall be able to provide an efficient service of equipment, and must have ample stocks of spare parts. The Contractor shall submit a guarantee from the supplier that will hold for a sufficient number of spare parts for the maintenance of the equipment. The Employer reserves his right to reject any manufacturer or agents who do not fulfill the above requirements. It is the responsibility of the Contractor to provide evidence that the equipment is in compliance with this specification and that the equipment will operate satisfactorily, under the installed conditions. All equipment offered shall comprise of a complete installation kit such as supports, base frame, bolts, gaskets, guards etc to the satisfaction of the Engineer. Contractor shall carry out all tests specified in this specification or in a relevant standard together with any test which might be requested by the Engineer. Furthermore, the Contractor shall provide installation, operation, and maintenance manuals of each piece of equipment. <u>Program of Works for Machinery</u> The program of works for machinery to be supplied and installed shall be divided into the following parts Part I: Manufacturing Period Design and manufacture all equipment to be supplied under the contract including Inspection and factory testing (FAT). Part II: Shipping Period
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	After successful FAT, delivery of all equipment from the factory to the Contractor's Site including all freight loading, off loading, customs duties and clearance. Part III: Erection Period Removal of all materials and equipment from off/on Site storage, delivery to erection project Site, and installation. Part IV: Site Testing All Mechanical equipment and machinery shall be tested in the presence of the Employer before being put into operation. Part VI: Running-in and Tests on Completion All installed Mechanical equipment and machinery shall be adjusted to comply with the operational requirements and actual conditions. Manuals and other documentation shall be provided by the Contractor. (Test on completion shall take place during the completion phase of the project). <u><i>Purchase of Equipment</i></u> Prior to the purchase of any equipment, the Contractor shall submit, for approval by the Engineer, a complete schedule, with the names and addresses of the manufacturers and their product information catalogs, diagrams, drawings and such other descriptive data as may be required by the Engineer. Approval of equipment, under this provision, shall not be constructed as authorizing any deviation from the specifications, unless the attention of the Employer has been directed to the specific deviations. The Contractor shall produce the Manufacturer's certificate and copies of the delivery documents, for all such equipment, as and when required by the Engineer. <u><i>Installation and Commissioning of Equipment</i></u> The Contractor will be fully responsible for the installation and commissioning of all equipment (Pumps, Screen, Crane, etc), accessories, valves, fittings, piping works, instrumentations, and electrical works. PUMPS • Prior to purchase, the Contractor shall submit the following information, for the pump sets, for subsequent approval by the Employer: • The performance curves, efficiency curves, and power demand curves (characteristics curves), and life cycle cost shall be
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	provided by the Contractor with a clear indicator of the capacity, efficiency, and power required of the pump with the specified head. • Construction and other technical details • Overall dimensions and operating weight • Manufacturer's performance guarantee certificate • <i>The pumps, installed in the disposal station, should be suitable for parallel operation in all respects. The pumps should be so selected that they are not overloaded when only one pump is operating and increased water flow is to be handled due to reduced system head.</i> • <i>No. of operational and standby pumps and their selection criteria.</i> • <i>The Contractor shall supply approved anti-vibration foundation material, for isolating the pump foundations from the building structure.</i> • <i>The pumps shall be furnished with pressure gauges on the suction and discharge sides.</i> <u>WASTEWATER CENTRIFUGAL PUMPS</u> • <i>All the sewage centrifugal non-clogging dry pit pumps shall be of single-stage, volute casing in rugged construction, end suction, and radial discharge pumps.</i> • <i>The centrifugal pumps shall be specifically designed to handle sewage water containing 150 mm diameter solids, rags, fibrous materials, grit, and other wastes. The minimum solid passing size is 150 mm for pump design.</i> • <i>The centrifugal sewage pumps shall be installed with a cardan shaft.</i> • <i>All the centrifugal pumps shall be designed to operate continuously at maximum efficiency under specified conditions consistent with good operating characteristics and high efficiency.</i> • <i>The pumps shall have continuously rising head characteristics from the design point to shut-off.</i> • <i>The pump shall be designed to permit maximum interchangeability of parts for assembly.</i> • <i>All casting and forging shall conform to their respective materials specification and shall be free from flaws and objectionable imperfections, machined true and in a good workmanship-like manner.</i> • <i>All accessories required for proper and safe operation shall be furnished with the pumps.</i> • <i>All incidental piping (including valves, and fittings) required for venting, sealing, and lubrication of the pump shall be furnished by the Contractor.</i> • <i>The pump motor shall be supplied with minimum IP protection of 55 and should be designed to operate at 3.3 kV (Voltage) and 50</i>
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	Hz. The motor should be capable to operate on Variable Frequency Drive (VFD). • The efficiency of the pump at the duty point should not be less than 85% as per ISO 9906:2012 - 1U. • The duty point of the pump shall be selected near the best efficiency point and shall not be deviated more than 10% with reference to the best efficiency point. • The speed of the pumps shall not be more than 600 rpm. • The performance of the pumps shall minimum comply with ISO 9906 1U (no negative tolerance is allowed). • All the pumps shall be suitable for wastewater temperatures 5~35 °C. • All the pumps with motors shall be suitable for an ambient temperature of 50 °C, relative humidity of 90 %, and altitude of 208 m above sea level. • All electric motors shall be a minimum of IE3 efficiency class. • The sewage pump dealer/supplier/manufacturer shall provide certificate from parent/principal manufacturer proving that they are authorized dealer/supplier/manufacturer, and their authorized dealer/supplier is capable to provide after sale service locally. • Principal manufacturer shall have certification equivalent or above ISO 9001:2015 (The scope of certification shall include but not limited to Development, Production, Marketing and Service of Pumps) • Manufacturer should have test bed to test all parameters at full design capacity of pump being supplied. <u>Maintenance Manual</u> Upon completion, the Contractor shall furnish the Employer with a manual containing all the following items: • Description of equipment • Full operation and maintenance instructions • Valves operation • Fault finding chart • Emergency procedures • Maintenance and service period • Lubricating instructions • Primary and secondary spares • <i>Drawing - size A3</i> <i>The manual shall be specially written in English and not a standard manufacturer manual unless approved by the Engineer.</i>
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	<i>All instructions shall be written into the manual with reference to the drawings.</i> <i>The work shall not be considered to be complete for purposes of taking over until such instructions and drawings have been supplied to the Engineer.</i> <u>General Requirements</u> • <i>All Pumps shall be new and manufactured within 1 year from the date of shipment.</i> • <i>Manufacture warranty minimum 5 years from the date of commissioning.</i> • <i>All materials used in pump manufacturing shall be compatible with the sewer water to be handled</i> • <i>This specification together with the bill of quantities defines the minimum requirements of Pumps, Drivers (electric motor), and accessories.</i> • <i>Each Pump shall be in completely assembled condition. Anchor bolts for all pumps shall be provided as per the specification.</i> • <i>All Pumps shall have stable performance characteristics requiring that at the rated speed over the whole flow range it shall have a continuously rising curve i.e. the developed head shall continuously increase with decreasing flow up to shut off head.</i> • <i>Pump-Motor assembly shall be capable of starting with full valve open condition.</i> • <i>The motor selected shall be to suit the pump's operation requirement at the complete performance curve, with a safety margin of a minimum of 10% at the full curve requirement.</i> • <i>The impeller diameter of the Centrifugal Pump corresponding to the operating condition shall preferably not exceed 95% of the maximum possible diameter of the impeller. Impellers with a minimum impeller diameter for the casing model shall not be used.</i> • <i>The pumping unit and auxiliary equipment shall be designed and constructed for safe operation at full load duty and continuous service. The pumps shall be of robust construction, and suitable for long periods of maintenance-free service (24 hrs/day). Units shall be of a design proven in similar service with an excellent track record.</i> • <i>All equipment shall be designed and manufactured so that no damage results from inland transportation and during storage, installation, and operation of equipment under the climatic conditions to which it will be subjected.</i> • <i>Pump and electric motor assembly shall be designed and manufactured in a way to permit ease of access during the inspection, maintenance, and repair.</i>
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	• Clamps, lifting lugs, and eyebolts of adequate capacity shall be provided to ensure safe and easy handling of the pumps and motors. • The pump shall be completely assembled at the manufacturer's shop and mating parts shall be marked for ease of assembly at the site. • Compliance with this specification shall not relieve the Contractor of the responsibility of furnishing equipment and accessories of proper design, materials, and workmanship to meet the specified operating conditions. • Quality Assurance Plan for the pumps shall be prepared by the Contractor and shall be submitted for approval by the Engineer. <u><i>Standards and Applicable Codes</i></u> The design, manufacture, assembly and testing of the pumps covered by the specification shall unless specifically stated otherwise conform to the latest editions of all standards and codes mentioned below. In case of conflict between two different classes of specifications, the requirements of the more stringent two shall govern. • ISO 5199 - Technical specifications for Centrifugal pumps. • BS EN ISO 5198 - Centrifugal, mixed flow and axial pumps. Code for hydraulic performance tests. • ISO 21940-11 - Mechanical vibration - Rotor balancing - Part 11: Procedures and tolerances for rotors with rigid behavior. • ISO 3661 - End-suction centrifugal pumps - Baseplate and installation dimensions • ISO 281 - Rolling bearings - Dynamic load ratings and rating life • ASTM A48 / A48M – 03 - Standard Specification for Gray Iron Castings • ISO 9906-3B - Rotodynamic pumps - Hydraulic performance acceptance tests. • JIS C 4003 - Thermal evaluation and classification of electrical insulation. • EN 12756: Mechanical seals. Principal dimensions, designation and material codes • EN 60034-1: Rotating electrical machines <u><i>Materials and Design</i></u> • The pump shall be a centrifugal volute pump design with Grey Cast Iron (EN-GJL-250) construction. • The pump shall have an axial suction port, a radial discharge port, and a vertical shaft. • The pump shaft shall be directly coupled with the electric motor by means of a Cardan shaft.
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	• Pump casing shall preferably have interchangeable wear rings. • The impellers shall be the open type with a smooth surface, multi-vanes radial flow impeller and shall be made of Super Duplex Stainless Steel dynamically balance to ensure smooth running. • The pump shaft shall be of stainless steel 1.4021 + QT 800 or equivalent as per approval of the Engineer. • Suitable shaft mechanical seals or gland packing shall be provided unless approved otherwise. The seal type and material shall be compatible with the sewer water. • Grease lubricated ball or other type bearings of heavy-duty shall be used on the shaft. • The pump casing bearing, shaft, impellers, and gaskets must be executed of materials suitable for a minimum of 3 years of continuous operation in the sewer water system. <u>Casing</u> • The casing shall be so proportioned that the velocity of liquid be reduced gradually and converted to pressure with minimum friction and wear. The surface finish should be smooth. The design pressure of the pump shall be greater than 1.5 times the suction pressure plus the shut-off head of the pump. It should be rigid and properly supported to resist misalignment and any distortion from pipe loads. • Replaceable type wear rings shall be provided. The rings shall be so fitted as to prevent rotation with the pump running. • The pump's suction and discharge nozzles shall be with flanges drilled as per ANSI/AWWA C110/A21.10 / ANSI B16.1 or internationally acceptable standard. <u>Impeller</u> • The impellers shall be the open type with a smooth surface, multi-vanes radial flow impeller. • The impeller shall be trimmed to a diameter no less than 5% below the maximum diameter capable of being installed in the selected pump casing. The selected trim shall position the pump's rated duty point between 80% and 110% of its Best Efficiency Point (BEP) at the trimmed diameter. • The impeller selected shall be at least 5% smaller in diameter than the maximum size that can be accommodated in the selected size of casing and shall have high efficiency and low NPSH requirement. • Rotor assembly including impeller, shaft etc. shall be statically and dynamically balanced as per ISO 21940-11 Mechanical vibration -
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	Rotor balancing - Part 11: Procedures and tolerances for rotors with rigid behavior. • The impeller shall be keyed to the shaft and proper lock nut provision shall be made to prevent loosening of the impeller in case of reverse rotation etc. <u>Shaft, Shaft Sleeves & Wear Ring Rings</u> The shaft shall preferably be machined from bar / forging and ground finished. It should be adequately sized so that the maximum deflection of the shaft at the seal face under the worst condition of loading shall not exceed 0.05mm. Shaft sleeves and wear rings shall be replaceable and with high surface hardened materials. The wear ring shall be properly secured to prevent rotation relative to pump shaft rotation. <u>Shaft Seals</u> • All centrifugal Pumps shall be with Stuffing Box Gland Packing or fitted with Mechanical shaft seals suitable for the sewer water. <u>Bearings</u> Anti-friction bearings shall be standard type and shall preferably be selected for minimum L-10 rating. The bearings should be selected to give 3 years (25,000 hrs) life with continuous operation at rated operating conditions as per American Bearing Manufacturers Association (ABMA). The rise in bearing grease temperature with continuous running of the pump shall be within the allowable limit. <u>Base Plate / Frame</u> Pump shall be mounted on rigid steel structure which shall be embedded in RCC foundation. Suitable holes shall be provided for grouting and bolting of pumps to the structure. These holes shall be so located that the base can be grouted in place without disturbing the pump. <u>Electric Motors</u> • The pump shall be driven by a vertical, fan-cooled, permanent magnet synchronous motor. • All motors shall unless otherwise stated be suitable for 3300 volts +/- 10%, 3 phase, 50 Hz and shall be controlled through variable frequency drive (VFD). The electric motor shall be compatible with VFD drive and fitted with suitable designed bearings.
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- *The motors shall be a minimum of IE3.*
- *The motor shall be IP55.*
- *The motor speed shall not exceed 600 rpm or less, 10 poles.*
- *The motor shall be face mounted squirrel cage, and suitable for an ambient temperature of 48 °C and RH 90%.*
- *The rated output of the motor shall be the maximum power required by the pump under the described conditions of head and discharge plus an allowance for loss of power in shaft etc. the motor shall be safe for complete curve with minimum of 10% FOS.*
- *Electrically driven pump shall unless otherwise stated be coupled via cardan shaft to the motor.*
- *Insulation shall be class F.*
- *All electric motors shall be fitted with Anti-condensation heaters.*

TESTING

General Quality Control Tests

The Contractor shall perform all the following general quality control tests and submit all test results to the Employer for approval:

Standard Running Test

The pump shall be operated from zero capacity to the maximum capacity, shown on the performance curve. Readings shall be taken at a minimum of 5 capacity points, including one point within 2% of the design capacity specified. The pump shall be operated at a speed within 5% of the design speed. The tests shall be conducted in accordance with the accepted practices at full speed and unless otherwise specified, the procedure and instruments used shall conform to the latest applicable standards of the Test Code of Centrifugal Pump.

Capacity Measurement Test

The discharge capacity of the pump shall be measured by means of a standard calibrated electromagnetic flowmeter, venturi tube, orifice plate or as per approval of the Engineer.

Head Measurement Test

For head measurement, calibrated bourdon or other gauges with equivalent accuracy and reliability shall be used. All gauges shall be calibrated before and after each series of tests.

Power Input Test

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	The power input to the pump shall be determined with a calibrated electric device energy analyzer/ electronic multimeter. Measurement of Speed The rotating speed of the pump shall be obtained by a hand counter, stroboscope lamp. <u>Hydrostatic Test</u> A standard hydrostatic test on the pump impeller housing shall be made at 1.5 times the shutoff head, developed by the pump, or at twice the rated head, whichever is greater. The hydrostatic test pressure shall be held for not less than 30 minutes after all leaks have been stopped. A comprehensive Quality Assurance Plan (QAP) shall be submitted by the Contractor. <u>Testing Certification</u> The Contractor shall obtain certificates of tests performed on such materials or goods by an agency approved by the Employer and shall certify that the materials or goods concerned have been tested under the requirements of the Specification and shall give the results of all the tests carried out. The Contractor shall provide adequate means of identifying the materials and goods delivered to the site with the corresponding certificates. <u>Material Test Certificate</u> Materials of the various pump components shall be tested in accordance with the relevant standards. Test certificates for these shall be furnished by the Contractor. <u>Visual Inspection</u> Pumps shall be offered for visual inspection by the Contractor before shipment. The components of the pumps shall not be painted before the inspection. <u>SUBMERSIBLE PUMP</u> <u>General</u> • Pumps supplied shall be capable of satisfying the performance requirements specified. • Pumps shall be quiet in operation and free from vibration. • Casings unless otherwise stated shall be of the best quality closed-grained cast iron, capable of withstanding all pressures that may be produced due to normal operating pressure and surges.
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	• <i>The pump shafts shall be of adequate size to avoid the possibility of fatigue failure and shall be shock and corrosion-resistant. The pump shafts and elements shall be adequately designed to assume rigid support of the impeller and to rotate without whip, vibration or undue deflection at all operating speeds and under all operating conditions.</i> • <i>The pump assembly, including the motor shall be of the design and production of only one manufacturer and shall be in full compliance with these specifications. All submersible pumps shall be complete with motor, bearings, couplings, valves, fittings and pressure gauges.</i> • <i>Pumps shall be vertical un-choke-able single-stage centrifugal type with radial or mixed flow impellers and shall be suitable for continuous running under completely submerged conditions.</i> • <i>The pump and motor shall be a combined unit and shall be totally submersible.</i> • <i>The impellers shall be designed to give non-overloading characteristics over the complete working range of the pump and shall be capable of conveying water with solids.</i> • <i>Casing with large-flow cross-section and replaceable wear ring where applicable.</i> • <i>Impeller with large, unobstructed flow ways to ensure non-clogging operation, minimum free sphere passage of 50 mm for small pumps with flow rate 50 m³/hr or less and 75 mm for large pumps.</i> • <i>Common motor/pump shaft supported in anti-friction bearings with grease and/or oil lubrication.</i> • <i>The submersible pumps shall be specifically designed to handle water with solids, fibrous materials, grit and other wastes.</i> • <i>Speed shall not exceed 2900 rpm.</i> • <i>The submersible pump shall be provided with a duct foot bend.</i> • <i>Suitable rated PVC insulated double-sheathed copper conductor flexible cable of required lengths shall be supplied along with pump units.</i> • <i>The length of the flexible cable shall be sufficient to terminate at the control panel/isolator without any joint.</i> • <i>The unit shall have lifting eyes for lowering and lifting the pump for safe handling.</i> • <i>The unit shall operate on 400 V +/- 10%, 50 Hz, 3Ph, AC electric supply.</i> • <i>The cable connection/entry shall be watertight and properly insulated.</i> • <i>All pumps shall be suitable for water temperatures of 0~40 oC.</i> • <i>The submersible pump shall be rated to operate at water depths up to 10 m.</i>
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	• The outlet connection of the pump shall be flanged. Submersible pumps shall be installed on a guide rail system to allow for the removal and installation of the pump without entering the wet well. Guide rails shall be used to direct the pump in proper alignment with the stationary discharge piping and shall be of a dual rail design. The guide rail assembly shall include all components necessary to provide a complete and fully functional assembly including: • Discharge base (duct foot bend). • Guide rails and supports. • Cast iron quick-release fitting shall be mounted to or integral to the pump volute. • Lifting chain. All the pumps with a weight greater than 30 kg, shall be provided with a lifting arrangement with chain block. The pumps shall be capable of working for long periods without cleaning or attention and special precautions shall be taken to avoid wear on working surfaces due to grit. Material The material of construction (MOC) shall be as under: • The pump casing shall be constructed of Cast iron. • Impeller shall be Super Duplex stainless steel. • Shaft shall be stainless steel. • The discharge base (duct foot bend) shall be constructed of cast iron. • Guide rails and supports shall be constructed of stainless steel SS316. • The lifting chain shall be stainless steel SS316. • Fastening components, material stainless steel SS316, A4. Electric Motor The motor shall have the following features: • Dry-running three-phase asynchronous motor in watertight casing according to IEC standard or equivalent. • Protection type IP 68. • Insulation class shall be F as per IEC or JIS C 4003 (Thermal evaluation and classification of electrical insulation). • Tandem mechanical shaft seal.
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	• The motor windings shall incorporate thermal switches as a safeguard against overheating. <u>Testing and Quality Control</u> Refer to the previous section of centrifugal pumps. <u>ASSEMBLY</u> The pumps shall be assembled completely in the shop to ensure correct fitting of all parts and shall be match marked before shipment, unless the pump is shipped completely assembled, to ensure correct assembly in the field. All the pump's tests shall be carried out by and at the expense of the Contractor to establish that the performance requirements of these Specifications and the Contractor's guarantees have been fulfilled. The pumps shall be tested in the manufacturer's shop and the performance tests shall be made with the entire pumping unit. The tests will be carried out in the presence of the inspector of the Employer or a third party if required by the Employer. The test shall cover: a) Determination of the total head. b) Determination of rate of water pumped. c) Measurement of input to the pump or output of the motor. d) Determination of pump efficiency. e) Preparation of characteristic curve showing pump efficiency, flow and head f) NPSH tests shall be conducted with water as the medium to demonstrate the ability of the pump to draw the water from the designed suction lift. <u>Preparation for Shipment</u> • All flanges, nozzles, and any other openings shall be blanked-off (capped or blind flanged). All untapped openings shall be provided with substantial wood or metal closure. The ends of the small piping may be sealed with tape. • The Pump shall be supplied with a Name plate as per relevant standards containing all the information required by the code.
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	However, the following information must be incorporated in the Name plate even if it is not required by the code: a. Manufacturer's Name b. Type and Serial No. c. Speed d. Kilowatt Rating e. Capacity, rated f. Head, rated g. Weight of equipment • The name plate shall be of stainless steel and relevant information shall be punched into the same. • All auxiliaries, dismantled and packed separately, shall be properly marked with Item No., for which it is intended, by means of a metal tag. • Pump with auxiliaries shall be suitably packed, fastened to avoid damage during transit, and crated for shipment and storage for a period of 6 months. Lifting, unpacking, and handling instructions shall be securely attached to the exterior of the largest packing in a well-marked, weatherproof container. The upright position of lifting points, gross weights, and dimensions shall be clearly marked on each package. Each Package shall clearly identify the equipment contained therein <u>PERFORMANCE WARRANTY</u> The equipment and its accessories shall be guaranteed for a period of 5 years warranty from the date of commissioning of the equipment. The Contractor shall also guarantee that the equipment will operate satisfactorily and the performance and the efficiencies of the equipment when operating under normal conditions shall not be less than the guaranteed values. The vendor shall further guarantee that during the guarantee period they will replace all defective materials, accessories, etc. furnished or installed under the contract, free of cost to the purchaser. The warranty for the replacement repaired work shall be 5 years from the date of such replacement repair. • MECHANICALLY RAKED BAR SCREENS DESCRIPTION
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	• The Contractor shall furnish all tools, equipment, materials, and supplies and shall perform all labor required to complete the work as indicated on the Drawings and specified herein. The mechanical bar rake coarse screens shall be provided with design features that incorporate chain-driven (multi-rake). For final pumping station, crane cable operated type mechanical screen may be considered. The Contractor shall comply with the design criteria provided in the ER. The Client and PMC shall decide the most appropriate type of screening mechanism based on local conditions, cost, quality, reliability, proven design, and low operation and maintenance cost. • If another type of bar screens (cleaning mechanism: reciprocating rake (with chain or hydraulic assembly), cable operated type for deep channels) are proposed to reduce the operation and maintenance cost, the Contractor must demonstrate in their Bid, through documentation and specifications, that the suggested equipment is proven and meets or exceeds the design, quality, and performance requirements outlined in the design criteria. • The Contractor shall furnish and install mechanically-raked bar screens and appurtenant work as shown, complete and operable, including screens, rakes, drives, cleaning arrangement, mounts, spare parts, and all other accessories and appurtenances, in accordance with the Tender Documents. • Screens shall be provided with waste storage containers and the sizes of containers shall be as per the dimensions of the screen for easy removal and cleaning operations. For final pumping station, disposal truck shall be considered. • The screens shall be installed in RCC channels. <u>GENERAL REQUIREMENTS</u> The Contractor shall furnish and install mechanically raked bar screens with chain-driven (multi-rake design) or reciprocating rake front-cleaned brush type with chain or hydraulic assembly. The system shall operate by means of a rake mechanism to elevate the debris along the upstream side of the bar screen from the channel invert to the discharge chute. The debris shall then be removed by a conveyor or directly stored in a waste container. The screen shall be of the front/back or combination cleaned type. Standby screens may be manually hand-cleaned bar screens, the existing screening system shall be used as a standby screen so the Contractor must provide all necessary arrangements to divert the sewage towards new or existing screening system. The existing screening system shall be updated by the Contractor, replacing with a new SS316 manual screen as per design criteria. All necessary piping, civil works, new manholes shall be included in the scope of supply.
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	A proportional weir in SS316 shall be installed after each screen to maintain the velocity in the channel during low flow conditions to avoid the settlement of solids at bottom of the screen channel. <u>CONSTRUCTION:</u> • The equipment to be furnished and installed shall be an automatic, self-cleaning front/back/combination cleaned type with a chain drive mechanism or a single reciprocating rake with chain or hydraulic assembly entering the bar rack from the upstream side. In case of chain drive, two heavy-duty chains shall be driven by a drive shaft through a gear motor. A system of a number of rakes shall be placed and installed with a chain and sprocket system. • Each rake shall contain multiple teeth and/or brushes, and be aligned with bar screens for the removal of accumulated screening material. A scraper, placed at the point of discharge, shall clean the screened material from the rakes and ejects the debris. The screen operation shall be controlled through a timer program (cycle on/off), and with a water level sensor. • The design of the screen shall ensure minimum maintenance requirements for items such as motors, drives, chains, rakes, and bearings. The equipment shall be readily adaptable for installation and operation in the RCC channel. If equipment that requires a change or modification of the design as shown is submitted and accepted, the Contractor shall prepare and submit for approval, drawings showing all the necessary changes. If approved by the Engineer, such changes shall be made at the Contractor's own expense and shall include all mechanical, structural, and electrical work. • A torque limiter shall be provided on the drive of the screen that must stop the screen in case of obstruction/blockage of the rake. • Moving parts of the screens shall be fitted with properly design safety guards as per guidelines of internationally acceptable standards. <u>Materials</u> • All materials used shall be suitable for service in a moist, corrosive environment as encountered in a Sewerage system or sewage disposal station. • Complete mechanical and manual bar rake screen (bars, screen frame, chain, sprocket, back plate, scraper, structural supports, etc) shall be made of stainless steel SS316. • Unless otherwise shown, all bolts, washers, nuts, and anchors shall be stainless steel SS316. The Contractor shall have the
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	manufacturer supply of initial lubrication and all necessary sealing fluids. <u>CONTROL SYSTEM</u> The mechanical screen system shall be furnished complete with an electrical control system. The control system shall include the following items: • The mechanical screen system shall be operated with water level sensors and a program time control shall also be designed for automatic cleaning frequency over a 24-hour period. • The screen shall be operated based on head loss, complete with a differential level indicator. The controller shall sense head loss through the screen. When a predetermined field adjustable head loss is reached, the differential controller overrides the time control and automatically starts the rake. • A local HAND-OFF-AUTO switch-activated alarm shall be reset by setting switch to "OFF." • A high differential water level alarm shall be provided and a dry contact shall be provided for the remote alarm in case of the SCADA system. • Control circuit breakers. • Obstruction Alarm and Contact: Obstruction shall stop the unit, a light shall be illuminated in the Control panel and a dry contact shall be provided to signal a remote alarm. Manually operated switch to reverse the direction of travel of the rake mechanism to clear the obstruction. <u>GENERAL INSTALLATION REQUIREMENTS</u> • Installation of the mechanically-raked bar screen shall be in strict accordance with the requirements of the manufacturer's written instructions and shop drawings; provided that nothing contained in said instructions or shop drawings shall authorize the Contractor to vary from the requirements of the Contract Documents. • All parts of the mechanism shall be designed in such a way as to avoid the development of stresses that may occur during fabrication, erection, and intermittent or continuous operation. Workmanship shall be of high grade in all respects and of the latest design. <u>FIELD TESTING</u> Upon completion of the installation, each piece of equipment and each system shall be tested for satisfactory operation without excessive noise, misalignment, vibration,
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	overheating, etc. All equipment shall be adjusted and checked for alignment, clearances, supports, and adherence to safety standards, until found satisfactory. <i>SCREEN CONTAINER</i> The screening waste container shall be a heavy-duty, corrosion-resistant unit, specifically designed for the collection, transportation, and secure handling of screening waste. The container body shall be constructed from high-quality Stainless Steel SS316, conforming to EN 10088-1 standards, ensuring durability and resistance to environmental conditions. Handles shall be made of Stainless Steel SS316 for ergonomic handling, while the container will be equipped with heavy-duty wheels featuring corrosion-resistant bearings to facilitate smooth mobility. A robust wheel locking mechanism shall be provided for stability and safety during stationary use. The container will also include a tamper-resistant locking system to prevent unauthorized access and ensure secure waste containment. The container shall be delivered ready for use, in accordance with these specifications and as shown on the Drawings or as directed by the Engineer. The container shall be resistant to wear and corrosion, and capable of withstanding regular handling in harsh conditions. Its construction ensures compliance with relevant standards, including ISO 9001 for quality management manufacturing firm. The design allows for easy access to critical components, such as wheels and locking mechanisms, for routine maintenance without disassembly. Upon installation, the container shall undergo functional testing to ensure proper operation of the locking and mobility mechanisms. The wheel locking system will be verified for stability, while the waste locking system will be tested for secure containment. The container will be measured and paid for on a per-unit basis, covering fabrication, material costs, transportation, installation, functional testing, and documentation. This ensures that the container is supplied, installed, and ready for operational use with minimal maintenance requirements over its service life. <i>ELECTRICAL OVERHEAD TRAVELLING CRANES</i> <u>GENERAL</u> This specification specifies requirements for an electric-operated overhead travelling crane. The crane should meet the requirements of internationally acceptable standards. The Contractor shall furnish and install EOT (Electric Operated Traveling) inside the new sewage pumping station at Gulshan-e-Ravi. This specification defines the minimum scope of the Contractor and requirements for design, engineering, fabrication, quality control documentation, inspection and testing at manufacturer premises, supply, transport, loading unloading, installation, testing and commissioning of the Overhead Traveling Cranes.
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	All necessary spare parts shall be included in the scope of the supply of Crane. <u>CODES AND STANDARDS</u> Latest editions with applicable amendment at the time of equipment ordering to be followed; • ASME B30.2: Overhead and Gantry Cranes (Top Running Bridge, Single or Multiple Girder, Top Running Trolley Hoist) • <i>EN 15011: Cranes - Bridge and gantry cranes or equivalent international standards like FEM, ASME, and DIN</i> • <i>ISO 4301-5: Cranes - Classification - Part 5: Overhead Travelling and Portal Bridge Cranes</i> • <i>BS 2853: Specification for the testing of steel overhead runway beams for hoist blocks</i> • <i>EN 14492-2: Power driven hoists.</i> • <i>EN 12385-4: Steel wire ropes-Safety</i> • <i>AWS D1.1: Structural Welding Code</i> • <i>IEC 60034: rotating electrical machines and LV motor data sheet.</i> <u>GENERAL REQUIREMENTS</u> • <i>The bridge crane system shall be controlled from a pendant push button station and shall include safety devices and overload protection. The rails shall be Standard I- beams of fabricated steel sections as indicated and as required by the relevant standard for seismic loads.</i> • <i>The track deflection shall not exceed 1/600 of the span. The Crane system except the tracks, shall be the product of crane Manufacturer regularly engaged in manufacture of such equipment.</i> • <i>The electrical operated overhead crane shall be minimum of Class D – Heavy service rated.</i> • <i>Complete inspection and testing plan shall be developed and submitted to Employer for approval. If there are any deviations, they shall be reviewed and approved by the Engineer. WPS/PQR/WPQ shall comply to AWS D1.1 / ASME Sec. IX. and all welders shall be qualified.</i> • <i>The main parts including hoist and girders shall be designed, fabricated, assembled and tested in the manufacturer premises.</i> • <i>For single girders the maximum factor span to safe working load shall not exceed 120t-m (Span x Safe Working Load).</i> <u>CONSTRUCTION REQUIREMENT</u>
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	<u>Hook and Wire Rope</u> The lifting hook shall be fabricated of drop forged heat treated steel shall include 360 degree Swivel on a shielded roller thrust bearing with safety spring latch. The wire rope shall be fabricated of Plow Steel. <u>HOIST AND DRIVE</u> The hoisting drum shall be deep grooved flanged drum with at least two full turns of rope remaining on the drum at lowest hook position and shall include heavy duty pre-lubricated sealed bearings. The drum shall be driven by a helical gear enclosed in an oil tight housing. The housing motor shall be a standard electric motor 1450 rpm. The Motor shall be rated at minimum of 150 percent of full load torque with gravity type upper and lower limit switches and an overload cut- off switch, designed to interrupt the raising circuit. <u>Trolley Assembly</u> The trolley assembly shall be a low/normal head room type. The trolley assembly shall be supported by trolley wheels with tapered tread hardened surfaces. Each wheel shall be supported on a tapered roller bearings suitable to take radial and thrust loads. The Wheel tread shall be smooth, true and uniform. <u>Trolley Drive</u> The trolley shall be driven by an electric motor with oil tight reducer gear motor conforming to NEMA specifications. The motor shall include cushion start and controller designed for smooth travel and load control. The driver shall provide synchronous drive from gear reducer to wheel. <u>Crane Bridge Assembly</u> The crane bridge assembly shall be a single beam/bridge or top running double beam center drive type. The bridge beam shall be designed in accordance with the specifications of Crane Manufacturer Association of America. It shall be fabricated of standard structural shapes complying with the AISC American Institute of Steel Construction Specifications. At Full load the beam shall be designed to limit the deflection to 1/600 of the span but no to exceed 1- ¼ inch maximum deflection. <u>End Trucks</u> The end trucks shall be traversed by stable assembly of structural shapes welded together to provide proper wheel and bearing alignment. The end truck wheel base shall be minimum of 1/7 of the crane span. One wheel of each end truck shall be geared and meshed with the pinion mounted on the crane squaring shaft. The end trucks shall be fastened to bridge beams with bolts to ensure alignment.
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	<u>Crane Wheels</u> The crane wheels shall have tread surfaces hardened to 375 to 425 Brinell. Treads shall be tapered to provide proper running alignment. Each wheel shall be supported on tapered roller bearings mounted on stationary axles, designed for radial and thrust loads. <u>Crane Drive</u> The Crane drive motor shall be totally enclosed and shall be integral with a fully enclosed oil splash lubricated gear reduction. The drive shaft and gear reduction shafts shall be supported by permanently lubricated precision ball or roller bearings. The drive shaft shall provide synchronous drive from the gear reduction to both end trucks. <u>Runway Beams and Rails</u> The runway beams and rails shall comply with the indicated requirements. The rails shall be an ASCE type securely fastened into the runway beams. The runway beams shall be designed from an ASTM A-36 structural steel shape and shall have a maximum deflection not to exceed 1/600 of the span. The beams shall be equipped with stops on both ends capable of withstanding the impact of fully loaded crane at 50 % of rated speed, and shall be field adjustable. Necessary column supports or clamps, hanger rod, bolts, and fitting shall be provided. <u>Electrical Controls</u> Electrical Controls shall be two speed as recommended by the manufacturer. Bridge control shall include a mainline magnetic contactor, manually operated fused mainline disconnect with lockout provisions, branch circuit fuses, reversing bridge control and transformer with fused secondary. Bridge control shall be mounted on bridge in an enclosure NEMA rated, actuated from a pendant pushbutton station suspended from movable trolley by means of a retractable cable to permit operation at 4 ft above all floor levels. Motors shall include cushion start. <u>Conductor and Wiring</u> The runway shall be provided with enclosed conductor based electrification. The Bridge shall have rigid truck festoon type electrification. Other wiring of the crane shall be in rigid or flexible conduit and in accordance with National Electrical code. <u>EXECUTION</u> <u>Installation</u>
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	Bridge crane shall be installed in accordance with the manufacturer's installation instructions, this section and requirements of the individual hoists and crane sections. <u>Field Testing</u> The CONTRACTOR shall field test all hoists and cranes to verify their rated – load Carrying capacity. <u>OPERATIONS AND MAINTENANCE MANUAL</u> Upon completion, the Contractor shall furnish the Employer with a manual containing all the following items. • Description of the equipment • Full Operations and Maintenance Instructions • Fault finding chart • Emergency procedures • Maintenance and Service Period • Lubricating Instructions • Primary and secondary spares • Spares for 5 years warranty period • Critical Spares • All Drawings size A3, one soft copy and 3 Hard Copies VENTILATION • Forced mechanical ventilation shall be provided where either safety, heat and/or odour problems are expected. Ventilation rate shall comply with national standards for health and safety and minimum of 10 air changes per hour. • Whenever forced ventilation is required the noise level shall be minimised and not exceed the levels as per • <i>All intakes and exhausts shall be covered for all weathers.</i> BEARINGS • All bearings shall be generously rated and sized to secure satisfactory and stable running without vibration under all conditions of operation for a minimum life of 50,000 hours running, which shall be certified or otherwise documented. • All bearing shall be to ISO standard and SI metric unit dimensions where practicable. • The intervals between lubrication shall be maximized (not less than 2 weeks) and shall be defined for each individual item and included in the operation and maintenance manual.
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	<i>DRIVE UNITS</i> • Drive units for the equipment shall be designed for 24 hours continuous operation. The nominal rating of the gear shall be at least equal to the nominal kilowatt output of the attached motor. Each gear shall be a totally enclosed unit with oil or grease lubricated antifriction rolling bearings throughout. • All drives, couplings and other moving parts shall be efficiently covered on all sides with safety guards. • Long-life seals on the input and output shafts shall be fitted up to prevent the escape of lubricant and the ingress of dust, sand and moisture. • Lubrication of bearings etc. shall be by either splash or forced feed system. • The Contractor shall ensure that the lubricants used for initial filling and specified in the maintenance manual is adequate for prolonged operation in ambient temperatures of up to 50 C without overheating. • The gearboxes shall be marked with manufacturer's identification together with the rated shaft speeds, output power and maximum ambient temperature. <i>PIPING WORKS</i> <u>PIPING MATERIAL AND CONNECTIONS</u> Piping material shall be ductile iron and conform to the codes and standards mentioned below. Piping work shall be carried out according to the requirements of this specification and bill of quantities and fabrication/installation drawings. Where specific details about the execution of any items of work are not included/specified, work shall be carried out according to the requirements of the latest editions of the following standards and Engineering's general specifications (after approval of the Engineer) described below: • ANSI / AWWA C151/A 21.51 Ductile Iron Pipe Centrifugally Cast • ANSI/AWWA C104/A21.4 Cement - mortar lining for ductile iron pipe and fittings • ANSI/AWWA C110/A21.10 Ductile and Cast-Iron fittings • ANSI/AWWA C111/A21.11 Rubber gasket joints for ductile iron pressure pipe and fittings • ANSI/AWWA C150/A21.50 Thickness design of ductile iron pipes • AWWA C500 Metal-Seated Gate Valves For Water Supply Service <u>PIPE AND PIPE FITTINGS</u>
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- The flanged pipe will be centrifugally cast ductile iron with smooth mortar linings as per AWWA C104. The pipe and flanges will have uniform thickness according to applicable codes and standards and have no casting defect or any other physical damage.
- Flanged elbow will be medium/long radius type. It will have smooth mortar lining in accordance with AWWA C104
- Dismantling joints shall be made of ductile iron and be approved for use in sewer handling systems and shall have an axial adjustment capability of at least 25 mm to facilitate easy dismantling and installation of pumps, valves, non-return valves, etc. Threaded rods and nuts through two opposite flanges shall secure the dismantling joint in the position determined during installation so that the system represents a rigid connection of two adjoining pipes. The material for hardware e.g., nuts, bolts etc. will be SS316.

STAINLESS STEEL PIPES

- Stainless Steel Pipes shall comply with EN 10217-7 and be accompanied by a test certificate according to EN 10204 / 3.1B. The certificates shall be part of the QA documentation.
- The pressure rating of the pipes shall be as determined in the drawings/Tender, however, at a minimum of PN 10.
- The dimensions of electric resistance welded or seamless pipes shall comply with ANSI B36.19 /EN 10220.
- Flanges for stainless steel pipes shall be executed as welding neck rings and loose flanges of similar quality as the piping.
- Stainless steel pipes shall be available with either welded or flanged joints, as specified. Welded joints shall follow appropriate welding standards, ensuring strength and leak-free performance. Flanged joints shall comply with ASME B16.5 specifications for appropriate pressure ratings.
- Gaskets in EPDM material for flanged joints shall be suitable for the working conditions and comply with DIN 2690/ISO-7483/EN 681-1 or equivalent. Bolts, nuts, and washers shall be stainless steel, complying with relevant standards for corrosion resistance.
- In addition, all stainless steel shall be accompanied by an authentic manufacturer's certificate to enable verification of grade, origin and date of manufacture.
- Welded stainless steel pipes shall have a wall thickness determined in the drawings, however of minimum as follows:

Internal diameter (mm)	Wall thickness (mm)
200 or less	2.0
250	2.5
300	3.0
400	3.0

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	<table border="1" data-bbox="646 329 1252 401"> <tr> <td>500</td><td>4.0</td></tr> <tr> <td>600</td><td>4.0</td></tr> </table> • All pipes, fittings, and joint materials shall be sourced from the same manufacturer to ensure compatibility and uniformity. • Pipes and fittings must be packed, transported, and stored in accordance with the manufacturer's recommendations to avoid damage. During transport, pipes shall be supported to prevent any deformation. Storage shall be on raised platforms, ensuring pipes are not in direct contact with the ground. • All pipes shall undergo tensile and hardness testing to ensure compliance with ASTM A312. • Pipes shall be hydrostatically tested per the standard to ensure they meet required pressure levels without leakage. • Pipes and fittings shall undergo visual inspections for surface defects, and dimensional tests shall ensure compliance with project requirements. GRP PIPES • GRP pipes for pressure applications shall comply with ISO 14692, with joints and fittings conforming to the same standard. The pressure class shall match the specifications in the drawings but not less than PN10 in all cases. GRP pipes must be UV-resistant and suitable for the site's environmental conditions, with storage and handling following the manufacturer's advice. • All pipes, fittings, and jointing materials shall be obtained from the same manufacturer to ensure compatibility and consistent quality. • Pipes and fittings must be packed, transported, and stored in accordance with the manufacturer's recommendations. Pipes should be supported adequately during transport to prevent deformation or damage. Storage on-site shall be on raised platforms or racks, with pipes secured to prevent rolling or contact with the ground. • Except for flanged joints or as approved by the Engineer, pipe joints shall be; • Flexible, sealed with rubber rings or gaskets, and capable of passing the required pressure tests. • Field-laminated joints, all procedures must comply with the manufacturer's guidelines. • Flanged connections shall be in accordance with ISO 14692 or BS EN 1092 standards, with proper gaskets and bolts as required. • Where mechanical couplings are used, they shall be suitable for the pressure rating of the system and certified for GRP.	500	4.0	600	4.0
500	4.0				
600	4.0				

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	• Suitable elastomeric gaskets in EPDM material shall be provided for all joint types and shall comply with DIN 2690/ISO-7483/EN 681-1 or equivalent. Gaskets must be compatible with the fluid being conveyed and resistant to aging. • For flanged joints, bolts and nuts shall be stainless steel A4, to ensure corrosion resistance as per specified. • Pipes shall undergo tensile and flexural strength tests in accordance with ISO 14692 to ensure structural integrity. • Each pipe shall be hydrostatically tested to the pressure specified in the standard or as required by the project. • Pipes shall undergo stiffness testing to confirm their ability to withstand external loads. • The manufacturer shall provide a certificate confirming that all GRP pipes and fittings meet the specified standards and project requirements. • Pipes and fittings shall undergo visual inspections for surface defects, and measurements of diameter, wall thickness, and other critical dimensions shall be carried out to ensure compliance with the project specifications. STEEL PIPES • The steel pipes shall not be used where the pipes are in direct contact with sewage wastewater and shall only be used where approved by the Engineer. • Steel pipes and fittings shall be factory-made. Factory welding of longitudinal and spiral seams shall be done by an automatic submerged/ arc welding process. The ends of pipes, fittings and specials shall be fully prepared for jointing by the specified method, prior to delivery to the site. • Steel pipes shall be manufactured and tested in accordance with DIN 1626 / API 5L grade B. • The dimensions of electric resistance welded or seamless pipes shall comply with ANSI B36.10 & 36.19 /EN 10220. • All units shall have a workmanlike finish, free from visual defects, and shall pass all specified tests. <i>External Coating</i> • External corrosion protection systems shall be applied at the factory, and pipes and fittings shall be supplied to the site with adequate quantities of • materials to permit the completion of the protection systems at welded joints on site.
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	• Surfaces to be protected shall be thoroughly degreased and cleaned to remove all grease, scale, rust or other extraneous matter. The surface shall be grid blasted to Sa 2½. • In buildings • Refer to the corrosion control section of this specification. • In the ground, outside buildings • Steel pipes shall be externally coated with a sprayed metallic zinc layer corresponding minimum 200g/m2 followed by a black bituminous paint not less than 25 microns thick applied at the factory. Special external bitumen-based hot-applied coating, conforming to BS 4147 and having a minimum thickness of 0.3 mm, shall be sprayed onto the pipes before laying to provide protection against corrosion. The coating shall be smooth, neither brittle when cold nor sticky when exposed to the sun, and shall be strongly adherent to the pipe. • Fittings shall be externally coated as specified above for pipes. • Internal coating like cement mortar lining as per AWWA C205, if required shall be as per Employer approval <i>Fittings</i> Steel fittings shall be factory-made and be in accordance with an appropriate international standard. <i>INSTALLATION OF PIPES</i> <i>General</i> Pipelines shall be level, plumb and properly aligned, respectively installed to the required slope. Utmost attention shall be given to all pipes connected to machinery and appliances for not transmitting forces and moments to the equipment. Forces and moments due to misalignment of pipe work shall under no circumstances be transmitted to duct walls. <i>Pipe Bends</i> Bends shall have minimum the same wall thickness and be of same material grade as the joining pipes. Radius 1.5 x D. <i>Tapers</i> • The tapers shall be of the concentric type, similar to DIN 2616. • Suction intake taper for pumps, however, shall be of the eccentric type. The installation of the taper for dry installed pumps shall be
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	with the straight side at the top. The length of the taper shall not be less than $0.5*(D+d)$. <u>Flush Appliances</u> The flush appliances shall be made of stainless steel. Each section, which can be separated from the overall piping system, by valves, must have a sufficient number of connections for drainage and cleaning. The exact number and location of connections shall be decided jointly with the Engineer. A flush connection comprises: • pipe socket DN 50 with outer thread 2", approx. 100 mm long • socket globe valve DN 50 • C-coupling with outer thread 2" according to DIN 14 307, material: brass • C-blind coupling according to DIN 14 311 complete with chain, material: brass <u>Vent Pipes</u> The vent pipes shall be made of stainless steel (EN 1.4401). Depending on the pipe arrangements, vent pipes including valves may become necessary. Generally, vent pipes must terminate in "safe" areas, e.g. above roofs. The arrangement of the pipes shall guarantee unobstructed ventilation. <u>Pipe Brackets</u> Pipe brackets shall be of stainless steel (EN 1.4401) AISI316 for all material of pipes in accordance with relevant section of this specification. Pipe bracket arrangements shall consist of sliding and fixed brackets. Design and installation of sliding brackets shall guarantee proper alignment of the pipe, even when the pipe is moving longitudinally. No stress shall be transmitted to the supports, except for the weight resulting from pipe and medium. The design of the fixed brackets shall be such that all forces and moments resulting from operating conditions are safely transmitted onto the building structure/RCC foundation. Required pipe supports, gussets, etc. shall be connected to the pipes by reinforced welds. The design of the brackets with respect to axial movement, static and dynamic forces e.g. from water hammer etc. shall be based on the prevailing operating and ambient temperatures of the pipe system in question. All brackets and pipe components being inaccessible after installation shall be sufficiently protected against corrosion prior to assembly.
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	<u>Manholes Covers</u> Covers of inspection manholes and street parcel manholes and frames shall be of ductile iron or composite material and rated as heavy duty. They shall comply with BS EN 124. The manholes shall be provided with reinforced concrete cover slab. Covers shall be securely lockable and complete with lifting keys. The clear opening shall be as per drawings but minimum of 600 mm. <u>Valve Surface Boxes</u> Surface boxes shall be of cast or ductile iron and shall be rated as heavy duty. They shall comply with BS 5834. Covers shall be lockable and secured to the frame when not in position. <u>TRANSPORTATION AND HANDLING</u> The Contractor shall provide protection for the ends of all pipes and fittings prior to the pipes and fittings leaving the place of manufacture in order to guard effectively against damage during transit and storage and the ingress of foreign matter inside the pipes and fittings. In handling pipes and fittings every care shall be taken to avoid distortion, flattening, denting, scouring or other damage. Pipes and fittings shall not be allowed to drop or strike objects and shall be lifted or lowered from one level to another by means of approved equipment only. The pipe shall be handled in such a manner that it is not pulled over sharp objects or cut by chokers or lifting equipment. Pipe and fittings that are damaged during transportation, handling or stockpiling shall be identified by Contractor, be informed to the Employer and proper remedial action should be taken in consultation with the Employer. <u>STORAGE AND SECURITY</u> • All pipes, fittings, valves, and appurtenances shall be stored at sites in or around Project area as approved by the Employer in accordance with the manufacturer's recommendations. • All piping material shall be stored by the Contractor in areas demarcated for this purpose. Pipes and fittings shall preferable be stored on dry wooden platforms. Pipe shall be stored on clean, level area to prevent undue scratching or gouging. • Valves, instruments, bolts, small fittings and gaskets shall be stored in covered, locked premises. In no case shall the marking on the equipment be removed. Valves and other equipment provided with protective caps shall be stored with the caps up to the moment they are installed.
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Lahore Wastewater and Drainage Management Section 6 – Employer Requirements (ER)
Project (LWDMP) – Sewerage System from Larechs
Colony to Gulshan-e-Ravi Lahore (Through
Trenchless Technology)

	• Contractor shall be entirely responsible for safekeeping of all the material and equipment. If any item is lost or damaged, the Contractor shall be required to replace it at his own expense. Contractor shall identify space for storage of material. • Pipe shall be stockpiled on level ground in such a manner as will prevent damage to any part of the pipe. During stacking and removal operations, safe access to the top of the stack is essential. Stacking types and the maximum stacking height shall be in accordance with manufacturer's recommendation. • If the pipe stacked for storage, such stacking shall be done in accordance with the pipe manufacturer's recommendations. • Pipes shall be supported adequately at all times and shall be stocked on a flat surface free of sharp objects. • Pipes shall be uniformly supported throughout the length. Alternatively timber supports of 3 inch wide shall be placed beneath the pipes with a spacing not greater than 1.5 meters. • Pipe stack height shall be kept to a minimum to ensure that the bottom pipes do not deform, and the stack shall be provided with side supports at spacing not greater than 1.5 meters. <u>TRACEABILITY</u> All piping material including pipes, fittings, flanges, bolting etc. shall bear marking as per their relevant manufacturing codes. The Contractor shall have Engineer's approval for all such materials before commencement of installation. All pipes shall be new and have manufacturers tests certificates of pressure and leak test, each traceable with its serial no. Each box, bundle or crate shall be legibly marked identifying the contents, and giving the name and address of manufacturer, name of the project, and date of dispatch. <u>VALVES</u> <u>GENERAL</u> • All valves shall be suitable for the service conditions under which they are required to operate. The design, construction and choice of materials shall take into account all operational deviations including pressure surge. • All valves with pumps shall be made of ductile iron. • All flanges of the valve shall be according to AWWA C110 / ANSI B16.1 class150. • The face-to-face dimensions of all flanged valves shall comply with AWWA C515 / C500 / ANSI B16.1 class150 or
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